# Machine-Checked Dimension Theory for Fusion Categories

A research proposal: pivotal and spherical structures, the double-dual twist, and a verified proof of Etingof–Nikshych–Ostrik Proposition 8.23

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**Abstract**

This proposal sets out a research programme whose central target is the first machine-checked proof of Etingof–Nikshych–Ostrik Proposition 8.23: *every pseudo-unitary fusion category admits a unique spherical structure whose dimensions are the Frobenius–Perron dimensions.* The theorem is the standard bridge from an analytic hypothesis (one scalar equation on the global dimension) to the algebraic structure that all of dimension theory presupposes, and it is used as a black box throughout the classification literature. No formal proof of it exists, and the categorical infrastructure on which it rests is only now arriving in Mathlib.

The programme builds on substantial preliminary work, reported in §4: a skeletal formalisation of multiplicity-free rigid fusion data with a proved inhabitant; a corrected duality normalisation that inhabits Fibonacci and Ising over $\mathbb{R}$; a reduction of the target to a single per-channel identity on the double-dual twist; a formalised proof of ENO Proposition 8.21; and a reconstruction of ENO's route through the square root $a$ of the quadruple-dual isomorphism, together with the identification of ENO's trace identity with multiplicativity of the dimension. One of these results is negative and changes the shape of the target: ENO 8.23 is *false* without its pseudo-unitarity hypothesis (a kernel-checked two-simple counterexample, the Yang–Lee category). A second witness shows that the dimension function depends on the choice of pivotal structure although the fusion ring is unchanged. The proposal turns these findings into a three-aim programme: a corrected and inhabitable formal signature together with its bridge to Mathlib's emerging pivotal and spherical category theory; a derivation of the double-dual twist from the duality data; and a formalisation of ENO's analytic core, yielding Proposition 8.23 in the scope actually formalised — multiplicity-free data over a field of characteristic zero, with the quadruple-dual theorem either taken as a hypothesis or proved by Bartlett's string-diagram argument.

The scope is narrower than the title of the source theorem and is fixed in §2.6: multiplicity-free data, a field of characteristic zero, and the quadruple-dual isomorphism either as a hypothesis or proved by Bartlett's string-diagram argument. The 24-month schedule is ordered by risk — corrected signature and exact target, reconstruction and trace correspondence, derived twist, ENO's normalisation — and the deliverables include an axiom-clean Lean 4 library, a verified negative-result note, upstream Mathlib contributions, and a documented methodology for auditing formal statements against their sources.

| **At a glance.** | |
|---|---|
| **Object** | Pivotal and spherical structures on fusion categories, and ENO Prop. 8.23. |
| **Method** | Skeletal finite data in Lean 4 (pinned to Mathlib `0df444a3`); exact arithmetic in $\mathbb{Z}[\sqrt{d}]$; kernel-checked counterexamples; axiom audit; bridge to Mathlib's categorical layer. |
| **Already done** | ∼5,200 lines, 23 modules; a green 3,032-job build at `d3d320e` (not yet a reproduced clean build); 73 audited declarations at `[propext, Classical.choice, Quot.sound]`; no `sorry`. |
| **Still open** | A corrected signature and an exact target; the twist derivation (the Ω formula in `PLAN.md` is retracted); complex scalars; the reconstruction and trace correspondence; ENO's $\sigma/\tau$ normalisation; the quadruple-dual theorem. |
| **Horizon** | 24 months, 7 work packages, 12 numbered deliverables. |

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# 1 Introduction

## 1.1 Dimension theory sits between decidable combinatorics and open structure

A fusion category over a field $\Bbbk$ is a $\Bbbk$-linear, semisimple, rigid monoidal category with finitely many simple objects and $\mathrm{End}(\mathbf{1}) = \Bbbk$. Its Grothendieck ring $K_0(\mathcal{C})$ is finite, decidable data: a finite set $I$ of simples, integers $N_{ij}^k = \dim \mathrm{Hom}(X_k, X_i \otimes X_j)$, a duality involution $i \mapsto i^*$. Almost everything one wants to compute about $\mathcal{C}$ — dimensions, the global dimension, Frobenius–Schur indicators, the $T$-matrix of a modular category, the state sum of a three-manifold — depends on more than this ring, but less than the full category.

The intermediate datum is a *pivotal structure*: a monoidal isomorphism $\gamma\colon \mathrm{Id} \to (-)^{**}$. It yields a dimension $\dim_\gamma(X) = \mathrm{tr}(\gamma_X)$ for every object, and a *spherical* structure — one with $\dim_\gamma(X) = \dim_\gamma(X^*)$ — is what makes the trace cyclic and therefore produces invariants of oriented manifolds rather than of framed ones.

Two facts locate the research problem precisely.

(a) *The dimension function is not a function of the fusion ring.* This is not a philosophical remark: the present project contains a kernel-checked witness (§4, `VecZ2Twisted`). The category $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ with non-trivial cocycle carries two pivotal structures whose dimensions on the non-trivial simple are $+1$ and $-1$ respectively. Any formalisation that treats "the dimension" as determined by the fusion rules is therefore wrong, and the first version of this project's pivotal axiom was wrong in exactly that way.

(b) *Whether a pivotal structure exists is open in general.* Etingof–Nikshych–Ostrik conjectured that every fusion category is pivotal (ENO, Conjecture 2.8); the spherical strengthening is stronger still. What is known is a sufficient condition of a different character: pseudo-unitarity.

## 1.2 The theorem that bridges the two

ENO Proposition 8.23 states that a pseudo-unitary fusion category — one whose categorical global dimension equals its Frobenius–Perron global dimension — admits a *unique* pivotal structure, which is spherical, and with respect to which the categorical dimensions of all simple objects are positive and equal to their Frobenius–Perron dimensions.

The proposition is used as a black box across the classification programme: in Ostrik's analysis of pivotal fusion categories of rank 3 and the resulting inequality, in the Haagerup–Izumi existence problem, in the passage from unitary anyon data to modular data, and in every place where one writes down a spherical category from a fusion ring plus positivity. It is, in a precise sense, the lemma that makes the rest of the subject work.

## 1.3 Why formalise, and why this is not transcription

A referee reading a two-page proof in a 2005 Annals paper cannot check, line by line, that the gauge chosen for the associator is the one in which the evaluation and coevaluation have the normalisation the argument assumes. A proof assistant can. The empirical case for this proposal is that when the present project first wrote down the statement, four defects surfaced, every one of them invisible to a reader checking that the formulas *looked* right:

**D1. The signature had no inhabitants.** The two unit axioms of the skeletal fusion datum were mutually inconsistent for $n \geq 2$: evaluated at $(0, b, 0, d, b, 0)$ they forced $1 = 0$. Every theorem quantified over the structure was therefore vacuous, including a "proof" of the target theorem.

**D2. The pentagon was not the pentagon.** The axiom called `pentagon` had eight free labels where the $F$-matrix pentagon requires ten, and it fails on $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ for *every* cocycle $\omega$, the trivial one included. A green build certified a false axiom.

**D3. The pivotal axiom fixed the twist to one.** `p_fusion` assumed $J \equiv 1$, where $J$ is the monoidal structure of $(-)^{**}$. With that assumption every fusion-rule table is pivotal via $p \equiv 1$, and the target theorem became provable with no hypothesis at all — the formal statement had drifted so far from the source that it asserted something false.

**D4. The second snake used the wrong inverse.** It wrote the right snake with a reciprocal of an associator *entry* where the $(0,0)$-entry of the *matrix inverse* of the associator block is required. The two agree for pointed categories and differ for Fibonacci, so the axiom excluded the two principal examples.

Each defect was found by making the statement inhabitable and checking it against the examples, not by re-reading it. This is the methodological thesis of the proposal: in this subject the value of formalisation lies less in the final certificate than in the audit that produces it, and the audit is transferable.

## 1.4 What has already been established

The preliminary results of §4 are not a plan; they are a library. In summary:

- a skeletal signature for multiplicity-free rigid fusion data, with a proved inhabitant ($\mathrm{Vec}_{\mathbb{Z}/2}$) and the pentagon checked on all 512 label tuples;
- a corrected duality normalisation, with the inverse associator carried as data, that inhabits Fibonacci and Ising over $\mathbb{R}$ — every field proved, including both snakes — and yields the spherical pivotal structure with dim = FPdim for both;
- a proof that a pivotal dimension is always a coboundary of the normalised double-dual twist, and that ENO 8.23 is *equivalent* to one per-channel identity on that twist;
- a formalised ENO Proposition 8.21: the spectral bound $|d_i| \leq \mathrm{FPdim}_i$ for any dimension function satisfying the eigenvector equation, together with the equality case;
- a kernel-checked refutation of the target as stated without hypotheses (Yang–Lee), and a second witness that the dimension function is not determined by the fusion ring;
- a reconstruction of ENO's own proof route — the square root $a$ of the quadruple-dual isomorphism, its monoidality defect $b$, and the trace identity — together with a proof that ENO's trace identity *is* the multiplicativity identity that the twist reduction isolates.

The library builds (3,032 jobs, warnings only), contains no `sorry`, and its axiom audit over 73 declarations reports `[propext, Classical.choice, Quot.sound]` throughout.

## 1.5 What is not yet established

The gap is now sharp, and it is not the algebra. It is the passage from a Hermitian structure on a pseudo-unitary category to positivity and multiplicativity of a chosen dimension function — ENO's analytic argument: Proposition 8.21 turns the *sum* equality $\sum_i |d_i|^2 = \sum_i \mathrm{FPdim}_i^2$ into a termwise equality $|d_i| = \mathrm{FPdim}_i$; a tensor automorphism of the identity, $\sigma = d^2/|d|^2$, then makes the dimensions real, and a *natural* automorphism $\tau = d/|d|$ — not monoidal in general — makes them positive. Alongside it, and logically prior to it, sit two defects of the signature rather than of the mathematics: the twist $J$ is carried as data constrained by four laws, not derived from $(\mathrm{ev}, \mathrm{coev}, F)$, and the scalars are real where the source theorem is over $\mathcal{C}$. Until $J$ is derived, the statement "$p$ is pivotal" mentions an object the mathematics does not fix.

The scope is narrower than the source theorem and is fixed here once (§2.6): multiplicity-free data; a field of characteristic zero; and the quadruple-dual isomorphism $g$ of ENO Theorem 2.6 either taken as a hypothesis or proved by the string-diagram argument of Bartlett §3.6. These are deliverables to be discharged, not silent assumptions.

### 1.6 Research questions

**Q1 (Signature).**
What is the weakest finite-data signature that is (i) inhabited by the standard examples, (ii) strong enough to express pivotality and sphericity faithfully, and (iii) provably equivalent to the abstract categorical notion?

**Q2 (Twist).**
Is the double-dual twist $J$ determined by $(F, \mathrm{ev}, \mathrm{coev})$? If so, what is its closed form, and does it satisfy the multiplicativity identity (M)?

**Q3 (Analytic core).**
Can ENO's proof of Proposition 8.23 be carried out inside the skeleton, taking as input only the pseudo-unitarity equation $\dim \mathcal{C} = \mathrm{FPdim}\mathcal{C}$?

**Q4 (Frontier).**
What is the largest statement provable *without* pseudo-unitarity — that is, what exactly is the content of ENO Conjecture 2.8 in a pivotal-free formalism, and which of its consequences survive as theorems about torsion in the universal grading group?

**Q5 (Method).**
Which failure modes of formalising monoidal category theory are systematic, and can they be detected mechanically rather than by luck?

### 1.7 Structure of the proposal

§2 fixes notation and states the target precisely. §3 surveys prior work, mathematical and formal. §4 reports the preliminary results in enough detail to be checked. §5 sets out the seven work packages and their milestones. §6 describes the techniques, including the ones that failed. §7–§9 give risks, timeline and deliverables. §10 argues for significance. Three appendices record the module inventory, the audit protocol, and the Lean statements of the main targets.

## 2 Background

### 2.1 Fusion categories and fusion rings

**Definition 2.1.** *A* fusion category *over a field $\Bbbk$ is a $\Bbbk$-linear abelian, semisimple, rigid monoidal category $\mathcal{C}$ with finitely many isomorphism classes of simple objects, such that* $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) \simeq \Bbbk$ *and the tensor product is bilinear and exact in each variable. We write $I$ for the set of isomorphism classes of simples, $X_i$ for a representative, $0 \in I$ for the class of $\mathbf{1}$, and $i \mapsto i^*$ for the duality involution determined by* $\mathrm{Hom}(X_i \otimes X_j, \mathbf{1}) \neq 0 \iff j = i^*$.

The Grothendieck ring $K_0(\mathcal{C})$ has $\mathbb{Z}$-basis $(X_i)$ and structure constants

$$X_i X_j = \sum_{k \in I} N_{ij}^k X_k, \qquad N_{ij}^k = \dim_{\Bbbk} \mathrm{Hom}(X_k, X_i \otimes X_j) \in \mathbb{N}.$$

In this proposal a *fusion ring* is the resulting abstract datum $R = (N, (-)^*, 0)$: the multiplicities, the duality involution, and the unit. The axioms used are those of a based ring: $N_{ij}^0 = \delta_{j,i^*}$; associativity $\sum_k N_{ij}^k N_{kl}^m = \sum_k N_{jl}^k N_{ik}^m$; and compatibility of the involution with the product. Commutativity is *not* assumed.

**Definition 2.2.** *A fusion ring is* multiplicity-free *if $N_{ij}^k \in \{0,1\}$ for all $i, j, k$. It is* pointed *if every simple object is invertible, i.e. $N_{ij}^k \in \{0,1\}$ and $\sum_k N_{ij}^k = 1$ for all $i, j$.*

Multiplicity-freeness is not a convenience in this project: with $N > 1$ the associator is a matrix between multi-dimensional Hom spaces, and a single scalar $F_d^{abc}(e, f)$ cannot hold it. The restriction is discussed in §6.4.

## 2.2 The Frobenius–Perron dimension

**Theorem 2.3** (Frobenius–Perron, for fusion rings)**.** *Let $R$ be a fusion ring with $n$ classes. There is a unique function $f\colon I \to \mathbb{R}_{>0}$ with $f_0 = 1$ and*

$$f_i f_j = \sum_k N_{ij}^k f_k \qquad \text{for all } i, j \in I. \tag{1}$$

*Moreover $f_{i^*} = f_i$, each $f_i$ is an algebraic integer, and if the fusion graph is connected then $f$ is the Perron–Frobenius eigenvector of every fusion matrix.*

The number $f_i$ is the *Frobenius–Perron dimension* $\mathrm{FPdim}(X_i)$, and $\mathrm{FPdim}(\mathcal{C}) = \sum_i f_i^2$ is the Frobenius–Perron global dimension. Note that (1) is the *character* form: $f$ is a ring homomorphism $K_0(\mathcal{C}) \to \mathbb{R}$.

In the present library Theorem 2.3 is not proved: the existence and uniqueness of $f$ is packaged as a structure `FPData`, supplied by hand for each example. Discharging it from Mathlib's Perron–Frobenius development is a deliverable (§5, WP0), not a research risk: the relevant material exists upstream as a chain of pull requests (§3).

## 2.3 Duality, pivotal structures, and the double-dual twist

Rigidity supplies, for each object $X$, an evaluation $\mathrm{ev}_X\colon X^* \otimes X \to \mathbf{1}$ and a coevaluation $\mathrm{coev}_X\colon \mathbf{1} \to X \otimes X^*$ satisfying the two snake equations. In a skeleton, both are scalars, and the composite

$$\kappa_i \;=\; \mathrm{ev}_i \cdot \mathrm{coev}_i \;\in\; \Bbbk^\times$$

is the loop scalar of $X_i$. It is *not* the dimension: the dimension depends on a choice of identification of $X$ with its double dual.

**Definition 2.4.** *A* pivotal structure *on $\mathcal{C}$ is a monoidal natural isomorphism $\gamma\colon \mathrm{Id}_{\mathcal{C}} \to (-)^{**}$. Its* dimension *is $\dim_\gamma(X) = \mathrm{tr}(\gamma_X\colon X \to X^{**})$, computed as the composite of $\gamma_X$ with the evaluation of the right dual. The structure is* spherical *if $\dim_\gamma(X) = \dim_\gamma(X^*)$ for every $X$.*

Two remarks are essential and both cost the project a rewrite.

*Remark* 2.5 (The twist). The functor $(-)^{**}$ is monoidal, and its monoidal structure is *not* the identity. Write $J_{ij}^k$ for the scalar of $X_i^{**} \otimes X_j^{**} \to (X_i \otimes X_j)^{**}$ on the channel $k$. Then a pivotal structure on a skeleton is exactly a family $(p_i)_{i \in I} \subset \Bbbk^\times$ with

$$p_0 = 1, \qquad p_k = J_{ij}^k \, p_i \, p_j \quad \text{whenever } N_{ij}^k > 0, \tag{2}$$

and the resulting dimension is

$$\dim_i \;=\; p_i \, \kappa_i. \tag{3}$$

Sphericity becomes

$$\dim_{i^*} = \dim_i \;\; \forall i \qquad \Longleftrightarrow \qquad \kappa_{i^*} = p_i^2 \, \kappa_i \;\; \forall i. \tag{4}$$

*Remark* 2.6 (The trap). Two statements are easily conflated, and the project conflated them once. (i) *Objectwise*, every simple object $V$ of a fusion category is isomorphic to its double dual $V^{**}$ (ENO Proposition 2.1): the right dual $V^*$ is the unique simple $X$ with $V \otimes X \ni \mathbf{1}$, the left dual ${}^*V$ is characterised the same way, and semisimplicity gives ${}^*V = V^*$, whence $V \simeq V^{**}$. Consequently a skeleton may always be chosen with $i^{**} = i$ on the nose, and the signature does not restrict generality by assuming it. (ii) *Monoidally*, the functor $(-)^{**}$ need not be isomorphic to the identity, and a monoidal such isomorphism is *precisely* a pivotal structure. Statement (i) fixes the objects and says nothing about the monoidal structure; that structure is the twist $J$, and it is why $J$ is data. An earlier version of this project built (ii) into the signature as $J \equiv 1$, which made every fusion-rule table pivotal and the target theorem true without hypotheses; the claim that a skeleton can always be chosen with $(-)^{**} = \mathrm{id}$ *monoidally* is false, and it is retracted in `PLAN.md`.

## 2.4 Pseudo-unitarity and the target

**Definition 2.7** (Squared norms; ENO §2.1, §8.4; EGNO Definition 9.4.4)**.** *For a simple $V$ and any isomorphism $a\colon V \to V^{**}$ (which exists by ENO Proposition 2.1), the* squared norm *is*

$$|V|^2 \;=\; \mathrm{Tr}_V(a)\,\mathrm{Tr}_{V^*}\big((a^{-1})^*\big) \;\in\; \Bbbk^\times,$$

*independent of $a$. The* global dimension *is* $\dim \mathcal{C} = \sum_i |X_i|^2$*, and $\mathcal{C}$ is* pseudo-unitary *if* $\dim \mathcal{C} = \mathrm{FPdim}\mathcal{C} = \sum_i f_i^2$.

Definition 2.7 is the non-circular one: it uses only the objectwise isomorphisms of ENO Proposition 2.1, not a pivotal structure, and is therefore available *before* the existence theorem that is to be proved. In a skeleton with a chosen $a_i$, one has $|X_i|^2 = \kappa_i\kappa_{i^*}$; the library proves the form-independence of this product as `dim_mul_dim_dual`, so pseudo-unitarity is the equation

$$\sum_{i\in I} \kappa_i\,\kappa_{i^*} \;=\; \sum_{i\in I} f_i^2 \tag{5}$$

with no pivotal structure in it. The library's `IsPseudoUnitary` is exactly (5).

**Theorem 2.8** (ENO Proposition 8.23; EGNO Proposition 9.5.1)**.** *Let $\mathcal{C}$ be a pseudo-unitary fusion category. Then $\mathcal{C}$ admits a pivotal structure $\gamma$; $\gamma$ is spherical;* $\dim_\gamma(X) > 0$ *for every $X \neq 0$;* $\dim_\gamma(X_i) = \mathrm{FPdim}(X_i)$ *for every simple $X_i$; and $\gamma$ is the unique such structure.*

The two supporting results that ENO use are:

**Theorem 2.9** (ENO Proposition 8.21)**.** *Let $d\colon I \to \mathbb{C}$ satisfy the eigenvector equation $\sum_j N_{ij}^k d_j = d_i d_k$ for all $i,k$. Then $|d_i| \le \mathrm{FPdim}_i$ for all $i$. If moreover $\sum_i |d_i|^2 = \sum_i \mathrm{FPdim}_i^2$, then $|d_i| = \mathrm{FPdim}_i$ for all $i$.*

**Theorem 2.10** (ENO Theorem 2.6, after Radford)**.** *For any fusion category there is a monoidal natural isomorphism $g\colon \mathrm{Id} \to (-)^{****}$.*

Theorem 2.10 is taken as a hypothesis in this project (§4): proving it requires the weak Hopf algebra machinery of ENO §4–5 and is explicitly out of scope. Conjecture 2.8 of ENO — that every fusion category admits a pivotal structure — is not attacked directly; WP4 in §5 explains what the programme can say about it.

## 2.5 The skeletal point of view and what it costs

The formalisation works with finite data over a field $\Bbbk$:

$$\big(N_{ij}^k,\; i^*,\; F_d^{abc}(e,f),\; F^{-1},\; \mathrm{ev}_i,\; \mathrm{coev}_i,\; J_{ij}^k\big)$$

subject to the pentagon, the unit normalisations, the two inverse laws for the associator, the two snake equations, and the twist laws. This is the "$F$-symbol" presentation familiar from the anyon-model literature, and it has three costs, each addressed by a work package.

(i) **No abstract coherence.** A skeletal datum is a category only after a reconstruction theorem. WP5 supplies the bridge to Mathlib's categorical definitions and deletes the vendored patch once the upstream pull requests land.

(ii) **Real scalars, not complex.** The examples and `ENO823Exact` are over $\mathbb{R}$; the source theorem is over $\mathcal{C}$, where a pseudo-unitary datum typically produces roots of unity. WP1 adds complex support. One restriction that looked like a cost is not one: by ENO Proposition 2.1 a skeleton always has $i^{**} = i$, so the signature does not restrict generality there (§2.3, Remark 2.6).

(iii) **Multiplicity-freeness.** The signature assumes $N \le 1$ and the programme does not relax it: §6.4 explains why a matrix-valued associator is a separate development rather than a tweak, and scope statement (S1) keeps it out of the 24-month deliverable.

### 2.6 The scope, stated once

The title of ENO Proposition 8.23 is unconditional; what this proposal can deliver in 24 months is not, and the difference is stated here so that no later claim can quietly exceed it.

(S1) **Multiplicity-free data.** The signature carries a scalar associator, so $N_{ij}^k \leq 1$. The general case needs matrix-valued associators and is a separate development (§6.4); it is a stretch goal reached through the categorical bridge, not a consequence of the multiplicity-free theorem.

(S2) **A field of characteristic zero, eventually $\mathbb{C}$.** The library is over $\mathbb{R}$; the theorem is over $\mathbb{C}$. The complex case is a deliverable (WP1), not an assumption: the hypotheses must be stated for a field containing the scalars that occur, and the instances must be re-verified there.

(S3) **The quadruple-dual isomorphism.** ENO Theorem 2.6 ($g\colon \mathrm{Id} \to (-)^{****}$) rests on the weak Hopf algebra theory of ENO §4–5, which is out of scope. Two honest options are scheduled: take $g$ as a hypothesis and state the theorem conditionally, or formalise Bartlett's string-diagram proof of the same result (Bartlett Theorem 3.28 and Corollary 3.30, §3.6), which avoids weak Hopf algebras entirely. The proposal prefers the second and budgets it; the fallback is the conditional statement, clearly labelled.

(S4) **What is excluded.** The weak Hopf algebra input; the general-multiplicity associator; the classification of rank-three pivotal categories beyond the inequality used as a test; and any claim about ENO Conjecture 2.8 beyond the partial statements listed in WP5.

A theorem proved under (S1)–(S3, first option) is *ENO 8.23 for multiplicity-free data, conditional on $g$*. The proposal says so on every statement that depends on it, and the deliverable list distinguishes the conditional and unconditional forms.

## 3 Prior work

### 3.1 The mathematical sources

**The source theorem.** The target is Etingof–Nikshych–Ostrik, *On fusion categories*, Annals of Mathematics **162** (2005), 581–642 [ENO]. The relevant material is §2.2 (the quadruple dual), §3 (the $T$-matrix and the bound), and §8.4 (pseudo-unitarity and the spherical structure). Two numbering conventions circulate: the published Annals version states the result as Proposition 8.23, while the arXiv version (`math/0203060v2`) states it as Proposition 8.22, one lower throughout §8.4. This proposal follows the Annals numbering and the library cites the number on every use; mis-numbering this proposition has already caused one wrong citation in the project's own plan.

**The textbook treatment.** EGNO, *Tensor Categories* [EGNO], states the theorem as Proposition 9.5.1 (its notes record that this is ENO 8.23), with pseudo-unitarity defined in Definition 9.4.4 and the global dimension defined through squared norms in §9.4 — the non-circular definition this proposal adopts in Definition 2.7. It is the more readable treatment, and it is the source of the convention that a pivotal structure is a monoidal isomorphism $\mathrm{Id} \to (-)^{**}$ rather than a choice of duals, which matters: the two formulations differ by a twist.

**An independent, coordinate-free proof.** Bartlett, *Fusion categories via string diagrams* [Bar15], gives a graphical development of exactly the results this project needs, and is the single most useful prior work for the proposed derivation of $J$:

- Theorem 3.28 and Corollary 3.30: the quadruple dual functor is monoidally isomorphic to the identity (ENO Theorem 2.6);
- Corollary 3.26: the coherence law for the pivotal symbols, in the form $\varepsilon^i_{jk,\alpha}\varepsilon^m_{kl,\beta} = \varepsilon^i_{nl,\gamma}\varepsilon^n_{jk,\delta}$ *whenever* $(F^i_{jkl})^{\gamma n\delta}_{\alpha m\beta} \neq 0$. The nonzero-entry qualification is part of the statement: without

it the F-free coherence law cannot be obtained by cancelling an associator entry that may vanish (§4.4);
- Theorem 4.17: an *explicit formula* for the pivotal operators $T^i_{jk}$ — the monoidal coherence of the canonical pivotal endofunctor — as the product of the *apex associator monodromy* and the *pivotal indicators*; this is the closed form the present project is missing for $J$;
- Theorem 5.4: pivotal structures on a (unitary) fusion category correspond exactly to tuples $(\gamma_i) \subset U(1)$ satisfying $\gamma_j\gamma_k = \gamma_i T^i_{jk}$ on every fusion channel, and sphericality is $\gamma_i = \pm 1$ for all $i$;
- Corollary 5.5: a pseudo-unitary fusion category admits a canonical spherical structure (the same statement as the target);
- §5.3: the *sphericalization* $\widetilde{\mathcal{C}}$ of $\mathcal{C}$ by the canonical $\mathbb{Z}/2$-action — an *equivariantization*, not a $\mathbb{Z}/2$-graded extension, and the correct description of the cover construction in WP5.

Theorem 5.4 is the cleanest available check on any derived formula for $J$: the two conventions are reciprocal (the present library's $p_k = J^k_{ij} p_i p_j$ against Bartlett's $\gamma_i = \gamma_j\gamma_k / T^i_{jk}$), and an index-level comparison gives an independent confirmation that does not go through the snake axioms at all. WP3 includes this comparison as an acceptance test.

**Applications and the shape of the literature.** Ostrik [Ost13] analyses pivotal fusion categories of rank three and derives the inequality

$$\sum_i d_i^2 \;\geq\; \sum_i \bigl|\textstyle\sum_j N^k_{ij} d_j\bigr|^2 / \ldots$$

that pins down the possible dimension functions; it is the place where "pivotal + positive + spherical" is used as a computational tool rather than as a structural hypothesis, and it is the template for the negative tests of WP6. The Haagerup–Izumi programme and the author's own work [Hua26] use the theorem at a different point, to convert coefficient identities (analytic) into a spherical structure (algebraic). Frobenius–Schur indicator theory [NS] is the other major consumer of the same layer: indicators are traces of pivotal structures composed with the duality, and they are generally *cyclotomic* integers — integrality of the higher indicators is a separate arithmetic statement, not a consequence of sphericity, and nothing in this proposal depends on it.

**What is *not* in the literature.** (a) No proof of ENO Conjecture 2.8 (every fusion category is pivotal); the spherical version is strictly stronger and equally open. The counterexamples to naive expectations are of two kinds: categories with no pivotal structure at all are not known, but categories where the dimension depends on the pivotal structure are *abundant* (the present project has a kernel-checked one, §4). (b) No computation of the double-dual twist $J$ in closed form for general multiplicity-free data; Bartlett's Theorem 4.17 is the closest, and it is stated through graphical monodromy rather than as an $F$-matrix formula ready for formalisation.

## 3.2 The formalisation landscape

**Mathlib.** `CategoryTheory.Monoidal.Rigid.Basic` provides `RigidCategory` (left and right rigid monoidal categories), with the file itself listing "define pivotal categories" as future work. There is no pivotal, spherical, or fusion category in Mathlib at the pinned revision, and no Perron–Frobenius theory for general irreducible non-negative matrices (only `Matrix.IsIrreducible`). The relevant upstream work is in flight:

- **PR #42145**, *the double right dual functor is monoidal* (+233/-4): defines `doubleRightDualFunctor` and its monoidal structure;
- **PR #42150**, *pivotal categories* (+370/-13 in 5 files): defines `PivotalCategory` as a rigid category with a monoidal natural isomorphism between the double dual functor and the identity; currently open and labelled *blocked-by-other-PR* on #42145;

- **PRs #42191** (spherical) and **#42192** (symmetric rigid is spherical), which build on #42150;
- the Perron–Frobenius chain **#39914**–**#39922**, whose declarations `PerronFrobenius.Irreducible`, `pft_irreducible`, `collatzWielandt` and `spectralRadius` supply the spectral theory the fusion-matrix argument needs.

The project's convention is to vendor a byte-compatible copy under `FusionCategory/MathlibPatches/` and to delete it, importing instead, when the pull requests merge. At the time of writing the vendored copies are *placeholders*, not code: they declare nothing that the project uses, `PivotalCategory.lean` is 33 lines and `PerronFrobenius.lean` is 22, and no theorem in the library depends on either. This is honest debt, and WP0/WP5 repay it.

Two claims about the library are kept apart throughout. *Green build*: `lake build` succeeds on the working tree at `d3d320e` (3,032 jobs, warnings only) and `AxiomAudit.lean` reports the permitted axioms on 73 declarations; this was observed. A *reproduced clean build* — a fresh clone, a fresh Mathlib cache, no path override — has not been observed, and is deliverable D2. Neither is evidence that the statements correspond to their sources; that is Gate 3 of Appendix B, and four failures of it are recorded in §4.

**Other formal and semi-formal resources.** Alphabeta Math's generated library page *Unitary Fusion Spaces, F/R Symbols, and Anyon Models* [AM] has multiplicity-aware fusion trees, the $F$-symbol as a recoupling matrix, the pentagon in fusion-tree coordinates, gauge covariance, and explicit Fibonacci and Ising carriers; it is pipeline-generated with provenance labels and is not a proof assistant development. TNLean [TN] is a blueprint project on tensor networks whose anyon chapter treats quantum-double models — a different object (microscopic lattice models rather than abstract fusion data). AnyonWiki [AW] supplies $F/R$/pivotal tables in a fixed gauge for multiplicity-free categories up to rank seven; it is *data*, not code, and the present project's Fibonacci and Ising conventions follow it rather than being derived afresh.

**Conclusion.** There is no formalisation of pivotal or spherical fusion categories anywhere, and the categorical infrastructure is landing in Mathlib during the period of this proposal. The strategic consequence is fixed: keep the skeletal layer aligned with #42150/#42191 naming, stub the bridge, and delete the patch when it merges. Reimplementing rigid category theory would be wasted effort; the value added here is the finite-data layer, the examples, the counterexamples, and the theorem.

### 3.3 The author's prior formalisation

The sibling package `haagerup-odd/` in the same repository is a completed formalisation of the main theorem of [Hua26]: `exists_EGEquations_odd`, the existence of the Evans–Gannon equations for the cyclic Haagerup–Izumi categories at every odd order. It comprises 115 modules and 24,295 lines, with 1,162 theorems and lemmas, no `sorry`, and an axiom audit reporting `[propext, Classical.choice, Quot.sound]`. It is pinned to the same toolchain as this project (Lean `v4.33.1`, Mathlib `0df444a3`).

Its relevance to this proposal is threefold. First, *feasibility*: the proposed library is roughly a fifth of that development in size, and the working method (finite data, exact arithmetic, per-file builds, an axiom audit in CI) is the one already proven at that scale. Second, *conventions*: both packages present fusion data as $F$-symbols with fixed gauges, and the Haagerup–Izumi coefficients are the natural large-scale instance for WP6. Third, *scope discipline*: that development stops at Chapter 9 of the source manuscript and excludes the categorical reconstruction, exactly as this one excludes the weak Hopf algebra input of ENO §4–5.

# 4 Preliminary results

This section reports what is already proved, in enough detail to be checked, and marks each claim with its epistemic status:

| Status | Meaning |
|---|---|
| kernel | proved in Lean 4; in `AxiomAudit.lean`; axioms within budget |
| kernel (witness) | a structure is exhibited and its fields are proved in Lean |
| script | verified by exhaustive computation in Python (`docs/*.py`), not formalised |
| conjecture | stated in `PLAN.md`, no proof |

## 4.1 The formal signature

The library separates the ring from the category. `FusionRing n` carries $N\colon \operatorname{Fin} n \to \operatorname{Fin} n \to \operatorname{Fin} n \to \mathbb{N}$, the involution `dual`, the unit at 0, associativity, and the two duality axioms. `FusionSystem n k` extends it with the categorical data:

| Field(s) | Content |
|---|---|
| `mult_free` | $N_{ij}^k \le 1$ |
| `F`, `F_support` | the associator $F_d^{abc}(e,f)$, vanishing off the channels |
| `pentagon` | the ten-label pentagon |
| `F_unit_left`, `F_unit_right` | unit normalisations at $a=1$ and $c=1$ |
| `Finv`, `Finv_support`, `Finv_comp`, `F_comp_inv` | the inverse associator and its two inverse laws |
| `ev`, `coev`, `ev_ne_zero`, `coev_ne_zero` | the duality scalars |
| `snake_left`, `snake_right` | the two snake equations |
| `J`, `J_ne_zero`, `J_unit_left`, `J_unit_right`, `J_dual`, `J_coherent` | the double-dual twist and its four laws |

A pivotal structure is then a family $p\colon \operatorname{Fin} n \to \Bbbk^\times$ with $p_0 = 1$ and $p_k = J_{ij}^k p_i p_j$ on every fusion channel; the dimension is $\dim_i = p_i \kappa_i$ with $\kappa_i = \mathrm{ev}_i \mathrm{coev}_i$; sphericity is $\dim_{i^*} = \dim_i$.

The one design decision that deserves comment is `Finv`. The associator at fixed $(a,b,c,d)$ is a square matrix whose rows are the channels of $b \otimes c$ and whose columns are the channels of $a \otimes b$; these are *different* subsets of $\operatorname{Fin} n$, so there is no canonical way to present the inverse as a matrix indexed by $\operatorname{Fin} n$. The project therefore carries $\alpha^{-1}$ as data with its two inverse laws. This is not pedantry: the right snake equation needs the entry $(\alpha^{-1})^{(0,0)}$, and getting this wrong excluded Fibonacci and Ising from the formalism (§4.3).

## 4.2 Four defects, and what caught them

| | Symptom | Diagnosis | Regression test |
|---|---|---|---|
| **D1** | every theorem vacuous | unit axioms inconsistent at $(0,b,0,d,b,0)$ for $n \ge 2$ | `VecZ2.system` |
| **D2** | "pentagon" fails on $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$, all $\omega$ | eight free labels instead of ten | 512-tuple check |
| **D3** | target provable with no hypothesis | `p_fusion` fixed $J \equiv 1$ | `VecZ2Twisted` witness with $\kappa_g = -1$ |
| **D4** | Fibonacci, Ising uninhabitable | reciprocal used for matrix inverse in `snake_right` | Fib, Ising |

D1–D3 share a signature: a statement that *looks* like its source but quantifies over an empty or over-constrained type. D4 is different — a genuine mathematical error in a normalisation — and it is the one that took three attempts (§4.3). The lesson recorded in the plan is that "the build is green" certifies nothing about the statement; only inhabitation and an independent computation do.

### 4.3 The duality normalisation

**Theorem 4.1** (Snake normalisation; kernel)**.** *Let* $\kappa_i = \mathrm{ev}_i \,\mathrm{coev}_i$*. In a skeletal rigid datum the two snake equations are equivalent to*

$$\kappa_i \; F_i^{i\,i^*\,i}(0,0) = 1 \qquad \textit{and} \qquad \kappa_i \; \big(F_{i^*}^{i^*\,i\,i^*}\big)^{-1}(0,0) = 1,$$

*where the second factor is the* $(0,0)$*-entry of the* matrix inverse *of the associator block, not the reciprocal of the single entry* $F_{i^*}^{i^*ii^*}(0,0)$*.*

The two readings agree exactly when the block is $1 \times 1$ — true for every pointed category, which is why the earlier "$\kappa/F$" correction sufficed for $\mathrm{Vec}^{\omega}_{\mathbb{Z}/3}$ — and differ as soon as a self-dual simple has a $2 \times 2$ block. Fibonacci is the minimal case: with the gauge $F_\tau^{\tau\tau\tau} = \left(\begin{smallmatrix} 1/\varphi & 1 \\ 1/\varphi & -1/\varphi \end{smallmatrix}\right)$ the left snake gives $\kappa_\tau = \varphi$, the reciprocal reading demands $\kappa_\tau = 1/\varphi$, and the matrix-inverse reading gives $\varphi \cdot (1/\varphi) = 1$ because the block is an involution. The library proves that the wrong reading fails (`Fibonacci.swapped_Finv_comp_ne_one`), so the correction is itself regression-tested.

Consequences, all kernel-checked:

- `FibonacciSystem : FusionSystem 2` and `IsingSystem : FusionSystem 3`, both over $\mathbb{R}$, with every field proved: pentagon, both unit laws, both inverse laws, both snakes;
- the pivotal structure $p \equiv 1$ is spherical for both, with $\dim = (1, \varphi)$ and $\dim = (1, \sqrt{2}, 1) = \mathrm{FPdim}$;
- $\mathrm{Vec}_{\mathbb{Z}/2}$ still inhabits the structure, so the correction did not buy the examples at the cost of the trivial case;
- the finite checks are done by *exact* arithmetic in $\mathbb{Z}$: each entry is an integer pair $(p, q)$ denoting $(p + q\sqrt{d})/2$ in $\mathbb{Q}(\sqrt{d})$, and `decide` verifies the identities. There is no `norm_num` over the reals and no `native_decide`.

The cost is compile time: the file takes about 100 seconds, so the package build is about three minutes.

*Remark* 4.2 (Statement of the snake in the general case)*.* The same computation in a non-skeletal category produces two extra structural maps (the duality comparison and the unit constraint). The skeletal form above is the image of the general snake under the equivalence, and the correspondence is the content of WP5's bridge; it is *not* obvious and should not be assumed silently.

### 4.4 The double-dual twist, and the two identities

**Theorem 4.3** (Pivotal dimension is a coboundary; kernel)**.** *Let* $P$ *be a pivotal structure and define the normalised twist*

$$\tau_{ab}^{c} \;=\; J_{ab}^{c} \cdot \frac{\kappa_c}{\kappa_a \kappa_b}.$$

*Then for every fusion channel,*

$$\dim_P(c) \;=\; \tau_{ab}^{c} \; \dim_P(a) \, \dim_P(b).$$

*Consequently* $\dim_P$ *is multiplicative on the fusion ring if and only if*

$$\sum_{c} N_{ab}^{c} \, J_{ab}^{c} \, \kappa_c \;=\; \kappa_a \kappa_b \qquad \textit{for all } a, b \qquad \textbf{\textit{(M)}} \tag{6}$$

*and ENO 8.23 is equivalent to the per-channel identity*

$$J_{ab}^{c} \;=\; \frac{f_c \, \kappa_a \kappa_b}{\kappa_c \, f_a f_b} \qquad \textit{on every channel with } N_{ab}^{c} > 0. \qquad \textbf{\textit{(K)}} \tag{7}$$

The equivalence is two theorems in the library, not one: `eno823_of_twistNorm` proves (K) ⇒ ENO 8.23, and `twistNorm_eq_fpdim_ratio_of_dim_eq` proves the converse. Together with `mult_iff_twist` ((M) ⇔ multiplicativity of $\dim_P$) they make the target a statement about $(N, J, \kappa)$ alone.

Two consequences deserve emphasis.

(i) **(K) is not invariant under the vertex gauge.** Rescaling $\mathrm{coev}_i \mapsto \lambda_i \mathrm{coev}_i$, $\mathrm{ev}_i \mapsto \mathrm{ev}_i/\lambda_i$ preserves the snakes and fixes $\kappa$, but multiplies the defect $\Delta^c_{ab} = J^c_{ab}\kappa_c f_a f_b/(f_c \kappa_a \kappa_b)$ by $t_a t_b/t_c$ with $t_i = \lambda_i/\lambda_{i^*}$. So (K), hence the target, survives only the duality normalisation $\lambda_{i^*} = \lambda_i$. The repair is to add $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ as an axiom, or to define $\dim_i = p_i \,\mathrm{coev}_i\, \mathrm{ev}_{i^*}$. This is a defect of the signature, not of the mathematics, and WP1 fixes it.

(ii) **On the coevaluation channel (K) is $J$-free.** On $(i, i^*; 1)$ the axiom $J^1_{ii^*} = 1$ applies, so (K) reads

$$\kappa_i \,\kappa_{i^*} = f_i^2 \quad \text{for all } i, \qquad \textbf{(C)} \tag{8}$$

which is exactly sphericity of $p = f/\kappa$. Hence (K) ⇒ (C) is a two-line theorem. The converse is *not* a theorem of the signature: (C) constrains $\kappa$ alone, while (K) constrains $J$ on every channel, and the pointed datum $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ satisfies the four twist laws and (C) and fails (K) (§4.5). What is true is that (C) is necessary for the target, and that (C) ⇒ (K) holds for the *genuine* $J$; supplying that implication — which is ENO 8.23 — is what the derivation of $J$ has to deliver. It is not a step that can be taken in the signature as it stands.

## 4.5 Refutations

**Theorem 4.4** (Yang–Lee; kernel)**.** *There is a complete skeletal rigid datum* `YangLee.system` *over* $\mathbb{R}$ *— Fibonacci's fusion ring, associator* $F_\tau^{\tau\tau\tau} = \left(\begin{smallmatrix} -\varphi & 1 \\ -\varphi & \varphi \end{smallmatrix}\right)$*,* $\mathrm{ev} \equiv 1$*,* $\mathrm{coev}_\tau = -1/\varphi$*, whence* $\kappa = (1, -1/\varphi)$ *— satisfying the pentagon on all* $2^9$ *label tuples, both unit laws, both inverse laws and both snakes, such that*

$$\neg\, \textit{ENO823 YangLee.system FibonacciFPData}.$$

*Indeed* $\kappa_\tau \kappa_{\tau^*} = 1/\varphi^2 \neq \varphi^2 = f_\tau^2$*, and on the channel* $(\tau, \tau; 1)$ *the axiom* $J^1_{\tau\tau} = 1$ *forces* $1 = \varphi^4$ *for any pivotal structure with* $\dim = \mathrm{FPdim}$*.*

Three things follow, and they shape the whole programme.

(a) ENO 8.23 is **false** as an implication from the axioms of a rigid fusion datum. The missing hypothesis is pseudo-unitarity, and no condition on $J$ can replace it: the contradiction uses only $J^1_{\tau\tau} = 1$, which is an axiom.

(b) The counterexample is *two-simple*. There is no hope that a rank threshold excludes it, and any formalisation that omits positivity will silently prove a false theorem — which is what the first version of this project did.

(c) Yang–Lee is not pseudo-unitary, which is exactly ENO's own remark. The formal statement `YangLee.not_pseudoUnitary` records the numerical failure $\sum_i \kappa_i \kappa_{i^*} = 1 + \varphi^{-2} \neq 1 + \varphi^2$.

**Theorem 4.5** (The dimension is not determined by the ring; kernel (witness))**.** $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ *with the non-trivial cocycle (*$F = -1$ *on the* $g \otimes g \otimes g \to g$ *block, duality normalisation* $\kappa_g = -1$*) is a* `FusionSystem 2` *over* $\mathbb{R}$*, carrying two pivotal structures:* $p \equiv 1$*, with* $\dim g = -1$*, and* $p_g = -1$*, with* $\dim g = 1$*, which is spherical. The conclusion of ENO 8.23 holds for the second and fails for the first.*

Theorem 4.5 is the counterexample to "the dimension is a function of the fusion ring". It is *not* a statement about the number of pivotal structures: both this datum and the untwisted $\mathrm{Vec}_{\mathbb{Z}/2}$ carry exactly two, and indeed once one pivotal structure exists the set of them is a torsor under the characters of the universal grading group (WP4), so the count cannot distinguish the two.

What the witness establishes is the dependence of the *dimension function* on the choice, and hence why pseudo-unitarity is the right hypothesis rather than a technical convenience: it selects, among the pivotal structures, the one with positive dimensions.

*Remark* 4.6 (A third witness, at script level). For $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $\omega = (-1)^{a_1 b_2 c_1}$ one has $\kappa = (1,1,1,-1)$, which is not a character of $(\mathbb{Z}/2)^2$. With $J \equiv 1$, (M) fails; with $J = \delta(\kappa) = \kappa_a\kappa_b/\kappa_c$ it holds. This computation is in `docs/pointed_twist_check.py` and is *not* yet kernel-checked. It is the reason the signature must not add $J^2 = 1$ as an axiom: for $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/3)^2}$ with $\omega = \zeta^{a_1 b_2 c_1}$ the honest twist squares to a primitive cube root of unity, and that category is unitary, so such an axiom would exclude a category to which the theorem applies.

The script's own census is weaker than it looks, and the proposal does not rely on it. The cyclic families $\mathrm{Vec}^{\omega}_{\mathbb{Z}/n}$ are run with a "carry cocycle" whose exponent is a multiple of $n$, so the cocycle is identically 1 up to floating-point error; and for cyclic groups in the standard gauge $\kappa$ is a character, so $J = \delta(\kappa) = 1$ and the census cannot separate the honest twist from the trivial one. The only data sets that test a non-trivial $\kappa$ are $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ and $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/3)^2}$, and neither is kernel-checked. Repairing this census is WP3 Task 3.4.

## 4.6 ENO Proposition 8.21

**Theorem 4.7** (ENO 8.21; kernel, modulo `DimEigen`). *Let $R$ be a fusion ring with `FPData` and let $d\colon I \to \mathbb{R}$ satisfy $\sum_j N_{ij}^k d_j = d_i d_k$ for all $i,k$. Then $|d_i| \le f_i$ for all $i$. If moreover $\sum_i |d_i|^2 = \sum_i f_i^2$ then $|d_i| = f_i$ for all $i$.*

The proof formalised is ENO's: the weighted norm $\|x\| = \sum_j f_j|x_j|$ is submultiplicative on the eigenvector equation, and `fpdim_char` — the fact that $f$ is a *character*, $f_i f_j = \sum_k N_{ij}^k f_k$ — bounds it by $f_i$. Two index conventions occur ($N_{ij}^k$ against $N_i^{jk}$), and the library proves the cyclic symmetry $N_{ij}^k = N_{jk^*}^{i^*}$ (the `cyclic1` statement) needed to pass between them (`cyclic`, `cyclic1`, `cyclic2`); the repo's axioms do not otherwise connect the second and third slots.

The equality case is `eq_abs_of_sum_sq`, and `abs_dim_le_fpdim_of_pseudoUnitary` discharges `DimEigen` from multiplicativity, so ENO's $|V|^2 = \mathrm{FPdim}(V)^2$ is available for any pivotal structure whose dimension is multiplicative.

*Remark* 4.8 (What 8.21 does not give). It is tempting to think that the $T$-matrix route bypasses multiplicativity. It does not. ENO's Remark 3.1 says that in the pivotal case the signs $\varepsilon_i$ and $b_j^{ik}$ are 1, so $T_j^{ik} = N_j^{ik}$ and the eigenvector equation $T_i d = d_i\, d$ *is* multiplicativity of the dimension. The library proves this identification (`tMatrix_mulVec`, `abs_tMatrix`: the domination $|(T_i)_{kj}| \le N_j^{ik}$ is an equality). The non-trivial $b$-symbols exist only when no pivotal structure is fixed, and their role is to prove $\dim \mathcal{C} \neq 0$ without one.

## 4.7 ENO's own route: the square root $a$ and the $b$-cochain

ENO's proof does not start from a pivotal structure — it produces one — so a signature that begins with $p$ cannot express it. The library therefore also carries ENO's input.

**Definition 4.9** (kernel, `SqrtData`). *A* square root of $g$ *is an additive isomorphism $a\colon \mathrm{Id} \to (-)^{**}$ with $a^2 = g$ monoidal, where $g\colon \mathrm{Id} \to (-)^{****}$ is ENO Theorem 2.6. In a skeleton, monoidality of $a^2$ has to be stated against $J^2$, because on labels $(-)^{**}$ is the identity: the tensorator of $(-)^{****} = (-)^{**} \circ (-)^{**}$ is $J^2$, not $J$.*

Given such an $a$, ENO set

$$b_c^{ij} \;=\; J_c^{ij}\,\frac{a_i a_j}{a_c}, \qquad T_c^{ij} \;=\; N_c^{ij}\,b_c^{ij}, \qquad d_i \;=\; \mathrm{Tr}(a_i) \;=\; a_i \,\mathrm{coev}_i\,\mathrm{ev}_{i^*}.$$

**Theorem 4.10** (kernel). *With these definitions:*

*(i)* $(b_c^{ij})^2 = 1$ *on every channel, and it follows from* $a^2 = g$ *monoidal, not from* $J^2 = 1$ *(`b_sq`);*
*(ii) the trace identity* $d_i d_j = \sum_c T_c^{ij} d_c$ *is* equivalent *to*

$$\kappa'_i \kappa'_j = \sum_c N_c^{ij}\, J_c^{ij}\, \kappa'_c, \qquad \kappa'_i = \mathrm{coev}_i\, \mathrm{ev}_{i^*},$$

*which is (M) with the duality normalisation* $\kappa'$ *(`trace_iff`);*
*(iii)* $d_i = a_i \kappa'_i$*, and* $d_i = a_i \kappa_i$ *only when* $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ *(`d_eq_kappa`).*

The content of Theorem 4.10 is an identification, not a discovery: ENO's "by the definition of $b$" step is (M). But the identification is what dissolves an apparent fork in this project's plan. "Derive $J$" and "assume multiplicativity" are not two research programmes; they are two faces of one theorem, and the condition $J$ has to satisfy is (M) itself.

*Remark* 4.11 (What must not be assumed). `Cover.lean`'s `S4Data.g_fusion` reads $g_c = g_i g_j$ on channels. That is the $J = 1$ special case and is wrong in general; it must be restated against $J^2$. Likewise $J^2 = 1$ and $b^2 = 1$ must not be added as axioms (§4.5).

## 4.8 The instances, and the audit

| Datum | Ring | $\mathbb{k}$ | $\kappa$ | Role |
|---|---|---|---|---|
| `VecZ2.system` | $\mathbb{Z}/2$ | $\mathbb{R}$ | $(1, 1)$ | inhabitant; pentagon on 512 tuples |
| `VecZ2Twisted.system` | $\mathbb{Z}/2$, $\omega \neq 1$ | $\mathbb{R}$ | $(1, -1)$ | two pivotal structures |
| `FibonacciSystem` | Fib | $\mathbb{R}$ | $(1, \varphi)$ | dim = FPdim, spherical |
| `IsingSystem` | Ising | $\mathbb{R}$ | $(1, \sqrt{2}, 1)$ | dim = FPdim, spherical |
| `k2210` | $K(1, 0, 1)$ | $\mathbb{R}$ | — | `FPData` certificate; FPdim $= (1, 2, 1)$, $D^2 = 6$ |
| `YangLee.system` | Fib | $\mathbb{R}$ | $(1, -1/\varphi)$ | refutes ENO823 without hypotheses |

`AxiomAudit.lean` runs `#print axioms` on 73 declarations covering every result quoted above and reports `[propext, Classical.choice, Quot.sound]` throughout. The package builds in 3,032 jobs with warnings only at the recorded commit. There is no `sorry` in the library (the single occurrence of the string is in a comment recording the axiom budget). Two limits of this evidence: the build is a green build of a working tree, not a reproduced clean build (§3), and an axiom audit certifies the *use of axioms*, not the *correspondence of statements with their sources* — the latter is Gate 3 of Appendix B, and it is where the defects of §4 were found.

`ENO823` itself is a `def`, not a `theorem`: it records the open target. The conditional theorem `eno823_of_pseudoUnitary` takes positivity and multiplicativity of a chosen pivotal dimension as hypotheses (`IsPseudoUnitaryPivotal`) together with `PFUnique`, and concludes the theorem. Both hypotheses are necessary: positivity fails on Yang–Lee, and multiplicativity fails on $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$.

## 4.9 The exact gap

Stated as a list, the gap between the present library and the target is:

(A) **The twist is data, not derived.** $J$ is constrained by four laws; the closed form carried in `PLAN.md` (the $\Omega$ transcription) is retracted there, and the census that was meant to support it is inconclusive (§4.5, Remark 2.5). So no correct derivation exists yet. Consequences: `p_mon` mentions an undetermined object, and (M) does not follow from the four twist laws as such — a datum may carry an arbitrary coherent $J$, and on $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ it fails — so (M) has to come from the derivation of $J$, not from the axioms.

(B) **The analytic step is absent, and its statement needs care.** Positivity is not formalised, and neither is ENO's two-stage normalisation: a *tensor* automorphism $\sigma = d^2/|d|^2$ of the identity, followed by a *natural* automorphism $\tau = d/|d|$ which is not monoidal. Neither is sign-valued a priori (WP3).

(C) **The signature is narrower than the mathematics.** Real rather than complex scalars; multiplicity-freeness assumed; the triangle axiom not stated (it is derivable, WP1); $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ not assumed, so (K) is gauge-fragile. The label question is *not* a restriction, by ENO Proposition 2.1 (§2.3).

(D) **The cover and grading layer is scaffolding.** `coverGlobalDim_double` is a reindexing identity on $\sum d^2$; `coverCanonicalPivotal` returns the input scalar with no `PivotalStructure`; the three former axioms of `Grading.lean` are gone and their content is a theorem plus two `def`s recording open targets.

(E) **No bridge to Mathlib, and no reconstruction.** `MathlibPatches/` contains placeholders, and `FPData` is supplied by hand. The skeletal datum has not been shown to be a fusion category, so every categorical statement about it — including "the categorical dimension" — is presently a statement about scalars.

These five items, together with the reconstruction that gives them meaning (WP2), are the work packages of §5.

# 5 Research programme

## 5.1 Aims, and the logical geography of the target

The programme has three aims, corresponding to Q1–Q3, and one exploratory strand for Q4 that is deliberately kept off the critical path.

**A1 (Signature and scope).**
A corrected signature and an *exact* target: complex scalars, the triangle law derived rather than assumed, the coherence law with the nonzero-entry qualification, a gauge-invariant duality normalisation, and a statement of ENO 8.23 whose hypotheses and conclusion match the source line by line. A1 also includes the reconstruction: the skeletal datum must be shown to *be* a fusion category before any categorical word in the statement is meaningful.

**A2 (Twist).**
The double-dual twist $J$ derived from $(F, \mathrm{ev}, \mathrm{coev})$ — not transcribed from a formula that `PLAN.md` has retracted — with its four laws and its qualified coherence proved, so that "$p$ is pivotal" becomes a well-posed predicate.

**A3 (Theorem).**
ENO 8.23 for multiplicity-free data: existence, sphericity, positivity, equality with FPdim, and uniqueness, conditional on $g$ or with $g$ proved by Bartlett's graphical argument, and stated over a field of characteristic zero.

**A4 (Frontier, exploratory).**
The pivotal cover as an equivariantization and the torsor of pivotal structures: statements about *differences* of pivotal structures, which is what the theory actually supplies. No existence criterion is claimed (§5.2).

The logical relations between the statements are worth recording once, because the project's own plan confused them for several weeks. Figure 1 is the map, corrected: the implication that is *missing* is the one that matters.

Three readings of the figure matter for planning. First, **no box in it is the theorem except (K)**, and (K) is a statement about a $J$ that has not yet been derived; the diagram is therefore a map of obligations, not of results. Second, **(C) and (M) are necessary and not sufficient**: each is satisfied by data on which the target fails, so neither can be used as a lemma toward the target in the signature as it stands. Third, **the derivation of $J$ proves nothing about the target by itself**; it removes an undefined object from the statement. This is why A2 and A3 are separate work packages and why neither can be skipped.

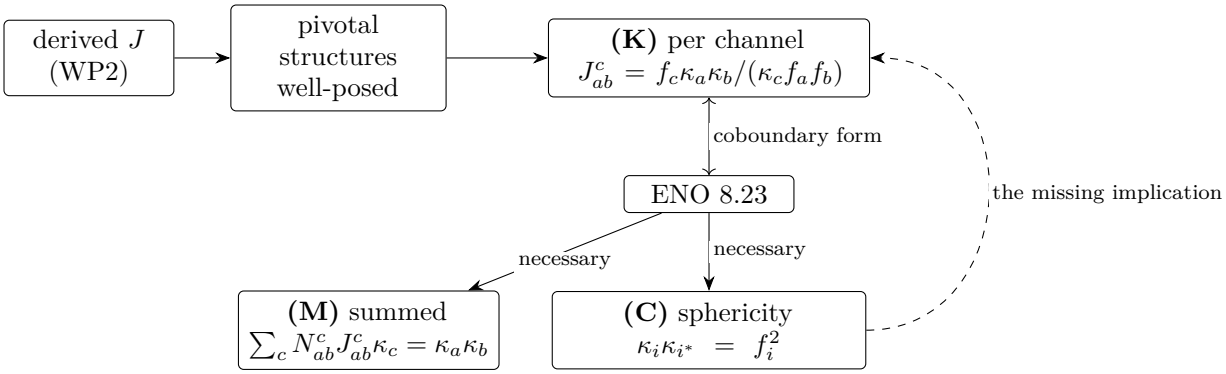



Figure 1: The logical geography, corrected. (K) is the coboundary form of the target and is equivalent to it in the signature. (C) is *necessary* but not sufficient: it constrains $\kappa$ alone, while (K) constrains $J$ on every channel, and the pointed datum $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ satisfies (C) and fails (K) (§4.5). The same holds for (M), which is weaker still and is satisfied by Yang–Lee. The dashed arrow (C)⇒(K) is *not* a theorem of the signature; it is part of the categorical argument that the reconstruction (WP2) and the derived twist (WP3) have to supply. The derivation of $J$ is logically prior: it makes the other boxes statements about determined objects.

## 5.2 Work packages

The order below is the order of the schedule and of the risk: nothing in WP3–WP4 is well-posed until WP1 has fixed the statement and WP2 has shown that the objects in it are categorical.

### WP0 — Foundations and reproducibility (months 0–4)

**Objective.** Replace hand-supplied inputs with proved or explicitly labelled ones, and make the build reproducible from a clean checkout.

**Tasks.**

0.1 Vendor the Perron–Frobenius chain (#39914–#39922) and define FPdim as the Perron root of the fusion matrix; prove Theorem 2.3 and delete `FPData` where it is not needed. Where uniqueness is genuinely required, keep the weaker `PFUnique` already in `PFDim.lean`; otherwise `FPData` stays an explicit hypothesis with a per-example certificate, and the scope statement (S2.6) says so.

0.2 Clean-tree CI: fresh clone, `lake exe cache get`, `lake build`, `lake env lean AxiomAudit.lean`, as separate gates. A from-clean build of the committed tree has never been verified — earlier builds used a path override — and that is a reproducibility debt no deliverable should inherit. This is also what turns the "green build" of §3 into a reproducible one.

0.3 A per-file build harness with a timeout, so that a 100-second example file does not block work on the core.

0.4 `BLUEPRINT.md` with the module dependency graph (Figure 2), the statement-to-source table of Appendix B (Gate 3), and the scope labels (S1)–(S4) attached to every theorem that depends on them.

**Deliverable.** D1, D2. **Success criterion.** A machine with the toolchain and no network builds the library and passes the audit from a clean clone.

### WP1 — The corrected signature and the exact target (months 1–8)

**Objective.** A1: make the finite-data signature express the mathematics, and make the formal statement of the theorem say exactly what ENO and EGNO say.

**Tasks.**

1.1 **Complex scalars.** Generalise `FusionSystem` and the target from $\mathbb{R}$ to a field $\Bbbk$ of characteristic zero (with the instances re-verified over $\mathbb{C}$). The source theorem is over $\mathbb{C}$; a real-valued formalisation is a restriction that must be stated, not hidden (§2.6, S2).

1.2 **The triangle law, derived.** The middle-unit entry $t = F_d^{a\,1\,c}(a, c)$ is *not* independent of the present axioms: the pentagon at $(a, 1, 1, c)$ gives $t = t^2$, and invertibility of the associator gives $t = 1$. Prove it as a lemma. The earlier plan to add it as a new axiom was wrong about its independence.

1.3 **Coherence with the nonzero-entry qualification.** The scalar coherence law may only cancel an associator entry that is nonzero; Bartlett's Corollary 3.26 states the pivotal-symbol coherence exactly that way. Restate `J_coherent` with the qualification (the signature already carries `F_support` and `J_ne_zero`, which supply it on the support) and prove that the F-free form follows from the qualified form under those hypotheses — rather than by cancelling a possibly vanishing entry.

1.4 **Gauge-invariant duality normalisation.** Either add $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ as an axiom, or define $\dim_i = p_i \,\mathrm{coev}_i \,\mathrm{ev}_{i^*}$ as ENO do. The first makes (K) gauge-invariant; the second matches the source but changes every instance proof. The proposal is to do the first and prove the two normalisations agree in the self-dual case.

1.5 **The exact target.** The library's `ENO823Exact` is

$$\texttt{IsPseudoUnitary}\ S \to \exists!\, P,\ P.\mathrm{IsSpherical} \wedge (\forall i,\ \dim_P i > 0) \wedge (\forall i,\ \dim_P i = f_i),$$

which does carry existence. Every statement *about* it must reproduce the existential conjunction verbatim; the earlier rendering of the target in this proposal split it into a spherical-structure field plus separate positivity and uniqueness fields, which is strictly weaker and is withdrawn (Appendix C). Prove the two forms equivalent, and state the conditional and unconditional versions separately.

1.6 **The label question, recorded as closed.** Double duality fixes isomorphism classes (ENO Proposition 2.1), so there is no label permutation to add. The plan to add one was based on the error corrected in Remark 2.6; the blueprint records the retraction and no work item remains.

**Deliverables.** D3. **Success criterion.** All six existing instances re-verified under the corrected signature, with the same axiom budget; the triangle law and the qualified coherence proved as lemmas, not assumed; a target statement that matches Theorem 2.8 field by field.

### WP2 — Reconstruction and the trace correspondence (months 3–10)

**Objective.** Make the words in the statement mean what they say. Until the skeletal datum is a category, "the categorical dimension" is a scalar defined by a formula, and the theorem is a statement about a formal system.

**Tasks.**

2.1 **A small reconstruction.** For multiplicity-free data, define the $\Bbbk$-linear category whose objects are formal finite direct sums of simples and whose morphisms are matrices indexed by channels; prove it is monoidal, rigid, semisimple, and that $\mathrm{End}(\mathbf{1}) = \Bbbk$. The associativity and unit constraints are the $F$-data and the unit laws; the snakes are the snake axioms.

This is the piece of the programme that converts a formal calculation into a theorem about fusion categories, and it is scheduled before the twist because the twist is a morphism of the reconstructed category.

2.2 **The trace correspondence.** Prove that the categorical trace of $p_i\colon X_i \to X_i^{**}$ is $p_i\kappa_i$, that $\mathrm{Tr}(a_i) = a_i\,\mathrm{coev}_i\,\mathrm{ev}_{i^*}$ is ENO's $d_i$, and that $\kappa_i\kappa_{i^*} = |X_i|^2$ in the sense of Definition 2.7. This is small, checkable, and closes the gap between the library's scalars and the source's dimensions.

2.3 **The twist as a morphism.** Identify $J_{ij}^k$ with the monoidal structure of $(-)^{**}$ on the reconstructed category — i.e. with a specific morphism $X_i^{**} \otimes X_j^{**} \to (X_i \otimes X_j)^{**}$ — so that WP3 derives a morphism and not a table.

**Deliverables.** D4. **Success criterion.** The reconstructed category is a `FusionCategory` in the sense of the library's own definition, and every scalar named in the target is the corresponding trace.

### WP3 — Deriving the double-dual twist (months 4–14)

**Objective.** A2: $J$ as a function of $(F, \mathrm{ev}, \mathrm{coev})$, derived and not transcribed.

**The retraction this package has to respect.** `PLAN.md` records that the displayed formula

$$\Omega(a,b;c) = F_0^{b^*a^*c}(c^*,b)\cdot(M_b)^{-1}(0,c)$$

is *defective as written*: read literally it returns $+1$ on $(\mathbb{Z}/2)^2$ with $\omega = (-1)^{a_1b_2c_1}$, where the honest twist is $\delta(\kappa)$, i.e. $-1$ on six channels. The construction of the comparison is right and the transcription is not. `docs/honest\_twist\_check.py` evaluates the twist from its defining property and never evaluates the displayed $\Omega$. No milestone below depends on the displayed formula.

**Tasks.**

3.1 **The comparison map.** Construct $\psi_{a,b}\colon (a\otimes b)^* \to b^*\otimes a^*$ as the unique map intertwining the evaluations, and prove it is an isomorphism.

3.2 **The ratio formula.** Prove

$$J_{ab}^c \;=\; \frac{\psi_{a,b}|_c}{\psi_{b^*,a^*}|_{c^*}},$$

and *then* derive a closed form in terms of $F$; if the closed form differs from the retracted $\Omega$, the retraction is the correct record and the new formula replaces it in `PLAN.md` and the blueprint.

3.3 **The four laws and the qualified coherence** of WP1 Task 1.3. This is the step that turns "$J$ is data" into "$J$ is computed", and it is independent of pseudo-unitarity.

3.4 **Repair the census.** The cyclic families in `docs/pointed\_twist\_check.py` use a "carry cocycle" whose exponent is a multiple of $n$, so it is identically 1, and for cyclic groups $\kappa$ is a character, so those cases cannot separate the honest twist from the trivial one (§4.5). Replace them with genuine cocycles: $(\mathbb{Z}/2)^2$, $(\mathbb{Z}/3)^2$, a non-trivial 3-cocycle on $\mathbb{Z}/3$, and a non-abelian group; kernel-check the two witnesses that are currently script-level.

3.5 **Independent confirmation.** Compare index by index with Bartlett's Theorem 4.17 and Corollary 3.26, and with the corrected census. A mismatch is a defect in the derivation, not in the check.

3.6 **(M) as an acceptance test.** For the genuine $J$, (M) should be a theorem — the dimension of a pivotal category is multiplicative. A failure of (M) after the derivation means the derivation is wrong; it is not evidence that (M) is an extra hypothesis.

**Deliverables.** D5. **Success criterion.** The library contains no hypothesis on $J$ beyond the four laws; they are theorems about the derived object; the census contains at least four genuinely non-trivial cocycles and is kernel-checked. **Explicit non-goal:** WP3 does not prove (K). The target is the strictly stronger per-channel identity, which fails for Yang–Lee and is WP4's content.

### WP4 — ENO's normalisation and the theorem (months 8–20)

**Objective.** A3: ENO 8.23, with the normalisation argument stated as ENO state it.

**Tasks.**

4.1 **Construct $a$.** An additive isomorphism $a\colon \mathrm{Id} \to (-)^{**}$ with $a^2 = g$ monoidal. `SqrtData` currently assumes $a$; this task constructs it from $g$ (Theorem 2.10), which is either a hypothesis (S3, first option) or a theorem of WP4 Task 4.2.

4.2 **The quadruple dual.** Either keep $g$ as an explicit hypothesis, or formalise Bartlett's Theorem 3.28 and Corollary 3.30: the quadruple-dual functor is monoidally isomorphic to the identity, proved by string diagrams (§3.6 of [Bar15]) and avoiding the weak Hopf algebra theory of ENO §4–5. This is the single largest prerequisite in the programme and is budgeted as such; if it slips, the theorem is stated conditionally.

4.3 **The bound.** From the trace identity $d_i d_j = \sum_c T_c^{ij} d_c$ with $T_c^{ij} = N_c^{ij} b_c^{ij}$ and $|b| = 1$, derive $|d_i| \le f_i$ by ENO's weighted norm. The library has the weighted-norm argument (`abs_le_fpdim_of_eigenvector`); what is missing is the passage from the $T$-matrix to the $N$-matrix, i.e. the triangle inequality $|\sum_j T_j^{ik} d_j| \le \sum_j N_j^{ik} |d_j|$.

4.4 **The sum identity.** Prove $\sum_i |d_i|^2 = \dim \mathcal{C}$ for $d_i = a_i \kappa_i'$, which under pseudo-unitarity gives $\sum_i |d_i|^2 = \sum_i f_i^2$; the equality case of Theorem 4.7 then yields $|d_i| = f_i$.

4.5 **The two normalisations, correctly typed.** This is the step the earlier draft of this proposal misstated, and it is worth stating exactly.

(a) The chain $f_i f_j = |d_i d_j| \le \sum_c N_{ij}^c f_c = f_i f_j$ is an equality; hence $T_{ij}^c = \pm N_{ij}^c$ and $d_i d_j / d_c$ is real whenever $N_{ij}^c > 0$. Consequently $\sigma_i = d_i^2 / |d_i|^2$ defines a *tensor automorphism of the identity functor*. It is not sign-valued a priori — it is a phase — and it is monoidal precisely because the ratios are positive.

(b) Twisting $g$ by $\sigma$ makes the new dimensions real. After this, $\tau_i = d_i / |d_i|$ is $\pm 1$; it is a *natural* automorphism, not a monoidal one, and it is applied to $a$, not to $g$.

(c) After the $\tau$-twist, $T = N$ and $b = 1$, so $a$ is a pivotal structure with $\dim = f > 0$.

The point to keep: multiplying $a$ by a monoidal character leaves the defect $b_c^{ij} = J_c^{ij} a_i a_j / a_c$ *unchanged*, so the $\sigma$-step cannot remove a defect; it normalises the phase. The $\tau$-step is the one that acts on $a$, and it is not monoidal. A draft that made both steps "tensor autoequivalences acting by signs" was wrong on both counts. (For a concrete illustration: in $\mathrm{Vec}_{\mathbb{Z}/3}$ the tuple $a = (1, -1, 1)$ satisfies $a^2 = 1 = g$ but is not a character, so rescaling a square root by arbitrary scalars changes the defect.)

4.6 **Assembly and uniqueness.** `eno823\_exact`: existence, positivity, equality with FPdim, uniqueness via `pivotal\_ext` and `PFUnique`.

**Deliverables.** D6. **Success criterion.** `ENO823Exact` proved for multiplicity-free data over a field of characteristic zero, with the hypothesis on $g$ either discharged (Bartlett route) or explicitly labelled; `YangLee.not\_pseudoUnitary` as the sharpness witness.

### WP5 — Cover, grading, and the frontier (months 14–22, off the critical path)

**Objective.** A4: the part of the theory that does not need pseudo-unitarity, stated precisely enough to be useful and no more.

**Tasks.**

5.1 **The cover as an equivariantization.** Bartlett §5.3: the sphericalization $\widetilde{\mathcal{C}}$ is the equivariantization of $\mathcal{C}$ with respect to the canonical $\mathbb{Z}/2$-action, *not* a $\mathbb{Z}/2$-graded extension of $\mathcal{C}$. Prove it is a fusion category, identify the forgetful functor, and prove $\dim \widetilde{\mathcal{C}} = 2 \dim \mathcal{C}$. The present `Cover.lean` is scaffolding: `coverGlobalDim\_double` is a reindexing identity and `coverCanonicalPivotal` returns a scalar with no `PivotalStructure`.

5.2 **The torsor.** Given one pivotal structure, the set of pivotal structures is a torsor under $\mathrm{Hom}(U_{\mathcal{C}}, \Bbbk^\times)$, where $U_{\mathcal{C}}$ is the universal grading group. This is a statement about *differences* between existing structures.

5.3 **What is withdrawn.** The draft claimed the criterion

$$\mathrm{Pivotal}(\mathcal{C}) \neq \varnothing \iff [g] \text{ is a square in } \mathrm{Hom}(U_{\mathcal{C}}, \Bbbk^\times),$$

and the corollary that $\mathcal{C}$ is pivotal when $U_{\mathcal{C}}$ has no 2-torsion. Neither is available. A map $g\colon \mathrm{Id} \to (-)^{****}$ is not an automorphism of Id, so its class is not a character class without an identification the project does not have; and the criterion fails even for monoidal $g$: $\mathrm{Vec}_{\mathbb{Z}/2}$ with $g$ the non-trivial character is pivotal, while that character has no square root in $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{C})$. A torsor does not establish existence. Any replacement needs a precise source and a separate proof, and is not promised.

**Deliverables.** D7. **Success criterion.** The cover is a `FusionSystem` with a `PivotalStructure` and is described as an equivariantization; the torsor statement is proved for differences; no existence criterion is stated.

### WP6 — The categorical bridge and upstream (months 12–24)

**Objective.** Mathlib's categorical definitions become the ones the library uses, and the vendored placeholders disappear.

**Tasks.**

6.1 **The pivotal equivalence.** On the category of WP2, prove PivotalStructure $S \simeq$ PivotalCategory (CategoryOf $S$): Mathlib's monoidal natural isomorphism $\mathrm{Id} \to (-)^{**}$ restricts to the scalars $p_i$ and `p\_mon` is its coherence.

6.2 **Delete the placeholders.** When #42145/#42150/#42191 merge, replace `PivotalCategory.lean` by an import. Until then the file declares nothing used elsewhere, and the blueprint says so.

6.3 **Contribute back.** The Perron–Frobenius material and the spherical layer are the two pieces of independent upstream value.

**Deliverables.** D8. **Success criterion.** `MathlibPatches/` contains no mathematics, only imports.

### WP7 — Instances, witnesses, and the regression suite (continuous)

**Objective.** Every theorem in the library has an inhabitant, and every defect in the table of §4.2 has a regression test.

**Tasks.**

7.1 Extend the instance set, over $\mathbb{C}$ as well as $\mathbb{R}$: $\mathrm{Vec}^{\omega}_{\mathbb{Z}/n}$ for $n = 2, \ldots, 6$ with genuine cocycles; Tambara–Yamagami for $\mathbb{Z}/2$; the rank-three ring $K(1, 0, 1)$ already present; $H_3$ (non-integer FPdim) as a test that nothing assumes integrality.

7.2 Formalise the two script-level witnesses of §4.5: $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ (the failure of (M)) and $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/3)^2}$ (the failure of $J^2 = 1$).

7.3 A *negative* multiplicity test. The based ring $X^2 = \mathbf{1} + 2X$ is legitimate, but it has no fusion-category categorification over $\mathbb{C}$ ([Ost02], Theorem 2.1). It is used to test the indexing of a matrix-valued associator and as a negative example against any claim that a based ring categorifies; it cannot supply a coherent rigid witness, and no such witness for $N = 2$ is promised by this programme.

7.4 A census harness with genuine cocycles, failing the build on disagreement with the reference computations.

**Deliverables.** D9. **Success criterion.** Sixteen data sets, at most two of them script-only, at least four with genuinely non-trivial cocycles, and the census harness green.

### 5.3 Milestones

| Milestone | Month | Content | Deliverable |
|---|---|---|---|
| M1 | 4 | `FPData` proved or labelled; clean-tree CI; blueprint with the statement-to-source table | D1, D2 |
| M2 | 8 | Signature v2 (complex scalars, triangle derived, qualified coherence, gauge fix); *exact* target; six instances re-verified | D3 |
| M3 | 10 | Reconstruction and trace correspondence; $J$ identified as a morphism | D4 |
| M4 | 14 | $J$ derived; four laws and qualified coherence proved; census repaired and kernel-checked | D5 |
| M5 | 18 | $a$ constructed; $T$-matrix domination; bound $\lvert d_i\rvert \le f_i$; sum identity and equality case | D6 (part) |
| M6 | 20 | $\sigma$ (tensor) and $\tau$ (natural) normalisations; `EN0823Exact` proved | D6 |
| M7 | 22 | Cover as equivariantization; torsor for differences; withdrawn criterion recorded | D7 |
| M8 | 24 | Bridge to `PivotalCategory`; placeholders deleted; release v1.0; papers submitted | D8, D10–D12 |

## 6 Techniques

### 6.1 Finite data and exact arithmetic

Everything categorical is reduced to finite algebra over a field, and everything numeric is done exactly. Two devices carry the instances.

**Integer pairs in $\mathbb{Z}[\sqrt{d}]$.** Real numbers never appear in a proof obligation. Each entry of $F$, ev, coev, $J$ in the Fibonacci, Ising and Yang–Lee examples is an integer pair $(p, q)$ denoting $(p + q\sqrt{d})/2$ in $\mathbb{Q}(\sqrt{d})$; the field operations are defined on pairs, and `decide` checks the pentagon, the unit laws, the inverse laws and the snakes on all label tuples. For Ising, $d = 2$ and the scalar $1/\sqrt{2}$ enters as a normalisation, so the identities are checked twice — once for the rational part and once for the $\sqrt{2}$ part. This is why the examples are trustworthy despite being numerical, and it is why they compile in 100 seconds rather than hours.

**Field checks are hand-written.** Closing real-number arithmetic is not automatic: `one_lt_pow` and its primed variant were unusable in the Yang–Lee file, and the $\varphi^4$ contradiction had to be assembled by hand from `nlinarith` with an explicit $\sqrt{5} \cdot \sqrt{5} = 5$. A mis-stated closed form for $((1 + \sqrt{5})/2)^2$ cost a build cycle. The practical rule adopted is: get the closed form right on paper, then prove it once, then reuse.

## 6.2 Inhabitant-first statement design

The four defects of §4 share one cause: a statement was written to look like its source and then trusted. The remedy, now institutionalised in this project, is that *no statement is committed until an inhabitant exists*, and no axiom is believed until an example satisfies it and a non-example does not. Concretely:

(i) every structure gets a witness in the same commit as its definition (`VecZ2.system` with the definition of `FusionSystem`);

(ii) every axiom gets a regression test that fails if the axiom is strengthened (`swapped_Finv_comp_ne_one` for the snake; `VecZ2Twisted` for the twist);

(iii) every theorem whose statement mentions a hypothesis gets a witness that the hypothesis is needed (Yang–Lee for positivity; $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ for multiplicativity).

The cost is real — the witness files are about half the library — and it is the single highest-value investment in the programme, because the defects it catches are exactly the ones that survive refereeing.

## 6.3 Falsification before proof

For each proposed statement $S$, the programme first looks for a counterexample, in this order: (i) on paper, on the smallest rank; (ii) in Python, on the full data set, with the associators composed as matrices rather than manipulated symbolically; (iii) in Lean, by attempting the statement on the counterexample candidate before attempting the theorem.

This order is not a heuristic. Two of the project's four defects were found this way, and a third — the '$J \equiv 1$' claim for pointed categories — was refuted by an exhaustive computation in `docs/pointed_twist_check.py` after it had been stated in the plan. The Python step matters because matrix composition over channel bases is where index conventions break, and because a script that recomputes every scalar from the snake axioms is an independent implementation, not a restatement.

## 6.4 Multiplicities: why this is a separate development

With $N_{ij}^k > 1$ the Hom space $\mathrm{Hom}(X_k, X_i \otimes X_j)$ has dimension $N_{ij}^k$ and the associator at fixed $(a, b, c, d)$ is a matrix whose entries are themselves matrices; the pentagon becomes a matrix identity with channel bases on both sides, and "the" $F$-symbol is a choice of bases. Three consequences:

(i) the scalar signature cannot be extended by adding a parameter — the type of `F` changes, and every instance proof that ends in `decide` over `Fin n` must be rewritten;

(ii) gauge freedom is larger (a $\mathrm{GL}_N$-valued change of channel bases); the twist $J$ becomes an operator between multiplicity spaces, and the condition it satisfies — multiplicativity, or the per-channel identity — is an *operator equality on each multiplicity space*, not a scalar equation. Taking a determinant or a trace of that equality destroys the monoidality information and does not give the scalar law: the general statement has no scalar reformulation of the kind the multiplicity-free case has;

(iii) the correct target in the general case is the *categorical* statement (WP6), with the skeletal statement as a corollary in the multiplicity-free case.

The proposal therefore treats the multiplicity-free theorem as the deliverable and the general case as a stretch goal, pursued through the categorical bridge rather than through a bigger scalar signature. This is a scope decision, stated here so that it can be argued with.

## 6.5 Diagram evaluation and independent reference computations

The one place where the project has repeatedly gone wrong is the evaluation of a diagram with more than two $F$-moves. The technique that works is to evaluate the same diagram twice by

different routes:

- an *evaluation-side* assembly (compose evaluations, transport coevaluations);
- a *coevaluation-side* assembly;
- the $\Omega$-coefficient form of WP2 Task 2.2;
- the direct $\varphi$-evaluation from its defining property $\mathrm{ev} \circ (\varphi \otimes \mathrm{id}) = \mathrm{ev}$, with full associator bookkeeping (`docs/honest_twist_check.py`).

Agreement of four independent assemblies on sixteen data sets is the current standard of evidence for a formula, and it is what caught the defective transcription of $\Omega$ recorded in the plan. It is *not* a proof, and the proposal does not treat it as one: the kernel proof of WP2 Task 2.3 is what settles the formula, and the scripts are demoted to regression tests.

A second, independent technique is the Temperley–Lieb check: in $\mathrm{TL}(\varphi)$ one can compute the loop scalar of Fibonacci with no $F$-data at all, and it gives $\kappa_\tau = \varphi$ against the naive $\varphi^2$. It certifies the snake normalisation and nothing else, which is exactly the scope claimed for it.

## 6.6 Parallel adversarial exploration

The project uses a workbench in which several agents attack one problem in independent directions, register their direction on a shared board, and submit reports that a coordinating agent accepts only after re-deriving or kernel-checking the claim. Two things are worth recording, because both bear on the proposed programme.

**What it bought.** The pointed-case value of the twist, $J = \delta(\kappa) = \kappa_a \kappa_b / \kappa_c$, was settled in this mode by three independent assemblies; the identification of the trace identity with (M) emerged from it; and four independent evaluations of the $J$-diagram agreed, which is what isolated the defective $\Omega$ transcription.

**What it got wrong.** The same mode produced claims that did not survive checking: that $X^{**} \cong X$ canonically in any rigid monoidal category (false — this is precisely pivotality, and adopting it would have made the signature vacuous, as it had already done once); a scalar formula for $\psi$ that returns 1 on channels where the honest twist is $-1$; and the assertion $J \equiv 1$ for pointed categories, which is false and was replaced by $\delta(\kappa)$. Each was caught by the acceptance rule, not by the generating agents.

**Consequence for the programme.** The technique is used for exploration and for independent assembly, never as evidence. Every claim in §4 marked "kernel" has been re-proved or re-checked by the coordinating agent; claims marked "script" are labelled as such; and no report is merged on the strength of its own text. The proposal's WP3 and WP4 explicitly budget for this: parallel exploration of the positivity step and of the cover, with kernel acceptance.

## 6.7 Statement fidelity as a method

The four defects are instances of four general failure modes, and naming them is part of the proposed contribution.

**F1. Vacuity.** The hypothesis is inconsistent, so the theorem is true and empty. Detected by inhabitation, not by proof.

**F2. Drift.** The formal statement is a different, usually stronger, statement (a one-line axiom where the source has a coherence condition). Detected by proving the source's examples fail, or by checking that the statement is not provable without its hypothesis.

**F3. Unrepresentability.** The signature cannot express the general case (real scalars where the source is complex; no multiplicity). Detected by trying to state the general theorem, not by proving the special one. Note that the label question is *not* an instance: double duality

always fixes isomorphism classes (ENO Proposition 2.1), and an earlier draft of this proposal wrongly listed $i^{**} \neq i$ here.

**F4. Normalisation.** The statement is right but the gauge is wrong: a reciprocal where a matrix inverse is required, a sign convention that excludes the examples. Detected by the examples, and only by the examples.

The audit protocol that follows is mechanical and is written up in Appendix B. It has three gates: *inhabitation* (does the structure have a witness?), *sharpness* (is the hypothesis used, i.e. does dropping it admit a counterexample?), and *correspondence* (is the statement the source's, with its numbering and its quantifiers?). The third gate is the one a proof assistant cannot automate, and the one that caught D3.

### 6.8 Engineering discipline

- **Pinned toolchain.** Lean `v4.33.1`, Mathlib `0df444a3`, in both packages; no bump without updating both lockfiles.
- **No `sorry`, no `axiom`**, axiom budget `[propext, Classical.choice, Quot.sound]` enforced by `AxiomAudit.lean`, which is a build gate and not a report.
- **Per-file builds** with generous timeouts (an `import Mathlib` costs 20–60 seconds; the example files cost minutes), `weakLeanArgs = ["-j2","-s8192"]` to stay inside CI memory limits.
- **No module-private declarations**: the project avoids the bare `module` header, which makes declarations private and breaks downstream imports in a way that is hard to diagnose.
- **Commit messages as a proof log**, including retractions. The repository's history carries three commits whose subject is a correction of the previous commit's mathematics; that is a feature, and the plan keeps the record rather than rewriting it.

## 7 Risks and mitigations

The risk of this programme is not concentrated in the analysis. It is concentrated in *statement correctness* and *reconstruction*: the preliminary work got four statements wrong before getting them right, and the analytic step cannot even be stated until the skeletal datum is a category. The table makes the dependencies explicit, in the order in which they bite.

| | Risk | Mitigation / fallback | Level |
|---|---|---|---|
| R1 | **Statement correctness.** A work package targets a statement that is not the source's, or a normalisation that is wrong. Four instances already occurred in the preliminary work, and an external reading of the first draft of this proposal found six more. | Gate 3 of Appendix B: every theorem carries its source, edition, numbering and quantifiers; the exact target is reproduced verbatim; two independent readings of each new statement before it is scheduled. | high |
| R2 | **Reconstruction.** The skeletal datum may not reconstruct as a fusion category from the present axioms, and the reconstruction is real category theory inside Mathlib. | WP2 does the multiplicity-free case first and is scheduled early; fallback: every theorem is labelled "skeletal" rather than "categorical" and the categorical claim is withdrawn from the abstract. | med/high |

| | Risk | Mitigation / fallback | Level |
|---|---|---|---|
| R3 | **The twist derivation.** No correct closed form is known; the $\Omega$ formula in `PLAN.md` is retracted, and the census that appeared to support it is inconclusive. | Derive through the comparison $\psi$, kernel-check, and cross-check against Bartlett's Theorem 4.17 and Corollary 3.26; fallback: keep $J$ a hypothesis and state the theorem conditionally on it. | med/high |
| R4 | **The quadruple dual.** ENO Theorem 2.6 needs weak Hopf algebras, out of scope; Bartlett's graphical proof is the alternative and is a large formalisation. | Budgeted as a separate task with its own deliverable; fallback is the conditional statement, labelled. | med |
| R5 | **Complex scalars.** The library is over $\mathbb{R}$; $\mathbb{C}$ breaks the `decide`-based instance proofs, which use integer pairs in $\mathbb{Z}[\sqrt{d}]$. | Model $\mathbb{Q}(\zeta)$ by integer polynomials mod $\Phi_n$ and keep `decide` exact; keep the $\mathbb{R}$ instances and add $\mathbb{C}$ ones rather than migrating. | med |
| R6 | **The normalisation step.** ENO's $\sigma/\tau$ argument needs the operator-level multiplicativity that WP2 supplies; if the reconstruction slips, this is unavailable. | The conditional theorem `eno823_of_pseudoUnitary` already isolates the gap; publish the conditional result if necessary. | med |
| R7 | Mathlib churn: #42150/#42191 merge with different names, or stall. | Vendor, keep byte-compatible, switch by replacing one import; contribute the spherical layer if it stalls. | low |
| R8 | Elaboration cost: `decide` over $3^n$ label tuples grows fast; example files already cost 100 s. | Exact arithmetic in $\mathbb{Z}[\sqrt{d}]$; per-file builds; abstraction barriers so the core does not import the examples; `native_decide` forbidden. | low |
| R9 | **Scope.** A single investigator over 24 months, on a theorem whose unconditional form exceeds the plan. | The scope statement (S2.6) is part of every deliverable; D1–D6 are the core and D7–D12 are additive; WP5 is off the critical path. | med |

**The stop conditions.** Two, and they are decided at fixed months rather than at the end. If by month 12 the reconstruction of WP2 has not produced a category, the programme drops the categorical claim and publishes the skeletal theorem as such. If by month 16 the twist derivation of WP3 has not succeeded, the theorem is stated with $J$ as an explicit hypothesis, and the deliverable is the conditional result together with the Yang–Lee sharpness theorem. Neither outcome is a failure; both must be decided on time.

## 8 Timeline

Twenty-four months, eight quarters. Bars mark active work. The order is forced by the dependencies: signature and exact target (WP1), then the reconstruction that makes the words mean something (WP2), then the twist (WP3), then ENO's normalisation (WP4). The cover and grading work (WP5) is deliberately late and off the critical path, because its statements are the ones most in need of justification.

| Work package | Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 |
|---|---|---|---|---|---|---|---|---|
| WP0 Foundations | ■ | ■ | | | | | | |
| WP1 Signature & exact target | ■ | ■ | ■ | | | | | |
| WP2 Reconstruction & trace | | ■ | ■ | ■ | | | | |
| WP3 Derived twist | | ■ | ■ | ■ | ■ | | | |
| WP4 ENO normalisation & theorem | | | ■ | ■ | ■ | ■ | ■ | |
| WP5 Cover & grading | | | | | ■ | ■ | ■ | ■ |
| WP6 Bridge & upstream | | | | | ■ | ■ | ■ | ■ |
| WP7 Instances & tests | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ |
| *Milestones* | | M1 | M2 | M3 | M4 | M5 | M6, M7, M8 | |

## 9 Expected deliverables

Twelve deliverables, each with a month and an acceptance criterion. D1–D6 are the core; D7–D12 are additive and independently valuable. The scope labels (S1)–(S4) of §2.6 apply to every deliverable below, and the conditional and unconditional forms of the theorem are listed separately.

| | Deliverable | Acceptance criterion | Month |
|---|---|---|---|
| D1 | **Core library v1.0**: fusion rings; Perron–Frobenius dimension with `FPData` proved or explicitly labelled; skeletal systems; pivotal and spherical layers; the twist reduction; ENO 8.21. | Reproducible clean build; no `sorry`; every audited declaration at `[propext, Classical.choice, Quot.sound]`. | 4–20 |
| D2 | **Reproducibility kit**: pinned toolchain, CI gates, clean-tree script, per-file harness, archived artifact with a DOI. | A fresh clone builds and audits without manual steps; the difference from the working-tree build is documented. | 4 |
| D3 | **Corrected signature and exact target**: complex scalars; triangle law derived; coherence with the nonzero-entry qualification; gauge-invariant duality normalisation; the target matching ENO 8.23 / EGNO 9.5.1 field by field. | All six existing instances re-verified; the new lemmas proved, not assumed; the target's existential conjunction reproduced verbatim. | 8 |
| D4 | **Reconstruction and trace correspondence**: the multiplicity-free datum is a fusion category; every scalar in the target is the corresponding categorical trace; $J$ identified as a morphism. | The reconstructed category satisfies the library's own `FusionCategory` predicate, and the trace identities are theorems. | 10 |
| D5 | **Derived twist**: the comparison $\psi$, the ratio formula, the four laws, the qualified coherence, a repaired census with genuine cocycles, and the Bartlett comparison. | No hypothesis on $J$ remains; at least four genuinely non-trivial cocycles, kernel-checked; the retracted $\Omega$ either corrected or replaced in `PLAN.md`. | 14 |
| D6 | **ENO 8.23**: $a$ constructed, $T$-matrix domination, the bound $\lvert d_i\rvert \le f_i$, the sum identity, the $\sigma/\tau$ normalisation, uniqueness. | `ENO823Exact` proved for multiplicity-free data over a characteristic-zero field; the hypothesis on $g$ either discharged (Bartlett) or explicitly labelled. | 20 |

| | Deliverable | Acceptance criterion | Month |
|---|---|---|---|
| D7 | **Cover and torsor**: the cover as an equivariantization, with doubled dimension; the torsor for differences of pivotal structures; the withdrawn existence criterion recorded. | The cover is a `FusionSystem` with a `PivotalStructure`; no existence criterion is claimed. | 22 |
| D8 | **Mathlib contributions**: the pivotal/spherical bridge; Perron–Frobenius dimension; the vendored placeholders deleted. | `MathlibPatches/` contains only imports. | 24 |
| D9 | **Instance and regression suite**: sixteen data sets over $\mathbb{R}$ and $\mathbb{C}$, at least four with genuinely non-trivial cocycles; the two script-level witnesses formalised; the rank-two negative test. | Census harness green; one regression test per defect D1–D4. | 18 |
| D10 | **Papers**: (a) the formalisation of ENO 8.23, with its scope in the title; (b) a negative-results note ("ENO 8.23 without pseudo-unitarity is false", with the kernel-checked Yang–Lee witness); (c) a methodology note on statement fidelity, including the corrections this revision records. | Submitted: (b) by month 12, (a) by month 21, (c) by month 24. | 12–24 |
| D11 | **Blueprint and documentation**: statement-to-source table, module dependency graph, gauge conventions, scope labels, a worked tutorial on adding an instance. | A reader can add a new instance without reading the core, and can tell which theorems are conditional. | 12 |
| D12 | **Open-problem dossier**: formal statements of ENO Conjecture 2.8 and its variants, with the known partial results and the constraints any counterexample must satisfy. | Reviewed against the literature; every claim labelled proved/checked/conjectured/resolved. | 24 |

**What is not a deliverable.** A proof of ENO Conjecture 2.8; the weak Hopf algebra input (ENO §4–5, Theorem 2.10) unless the Bartlett route succeeds; the general multiplicity case; the complex case beyond characteristic zero in the stated sense; and the classification of rank-three pivotal categories beyond the inequality used as a test. These are out of scope and are stated as such in the blueprint, so that a reader cannot mistake the boundary of the result.

## 10 Significance

**For the classification programme.** Proposition 8.23 is used as a black box in a large and growing literature; a kernel-checked proof would let downstream work cite a machine-checked theorem instead of a two-page argument whose normalisations are easy to get wrong. The present project's own history is the evidence: four defects, two of them in axioms, one of them turning the

target into a tautology. A reader who trusts the literature's normalisations is trusting something that a careful formaliser, working from the source, got wrong four times.

**For the mathematics.** The formalisation has already produced statements the literature leaves implicit: that (K), the per-channel identity, is *not* invariant under the vertex gauge and survives only the duality normalisation $\lambda_{i^*} = \lambda_i$; that (C) and (M) are necessary but not sufficient in the signature, and that both are satisfied by data on which the target fails; that the trace identity of ENO's proof is exactly (M); and that the coherence law needs the nonzero-entry qualification of Bartlett's Corollary 3.26. None of these is a new theorem of fusion categories in the sense of a new mathematical result, but each is a precision about hypotheses that the informal literature elides, and precision of this kind is what makes a black box safe to use.

**For Mathlib and the formalisation community.** The categorical layer is arriving now (#42145, #42150, #42191, #42192). What is missing after it lands is exactly what this proposal builds: the finite-data layer, the examples, the counterexamples, and one substantial theorem that exercises the layer. The Perron–Frobenius material and the spherical layer are independent upstream contributions; the sixteen verified data sets are the regression suite that the categorical definitions will need.

**For the author's programme.** The sibling Haagerup–Izumi development formalises the analytic side of a classification theorem (coefficient identities, reduction to finitely many algebraic equations). This proposal formalises the algebraic side of the same story (dimensions, pivotal structures, sphericity). Together they cover the two halves of a verified pipeline from fusion rules to a categorical structure, and the Haagerup–Izumi categories are the natural large-scale instance for WP6.

**Broader impact.** Verified anyon data — $F$ and $R$ symbols with a checked pentagon and checked pivotal structure — is the input to proposals for topological quantum computation, and the failure mode there is precisely an unchecked normalisation. The library does not build a quantum computer, but it does make the statement "these symbols satisfy the axioms" a machine-checked one, and that is the part which is currently done by hand.

**Limitations, stated plainly.** The theorem is formalised for multiplicity-free data in a skeleton; the general case is a stretch goal through the categorical bridge. The analytic input of ENO is formalised, but its weak Hopf algebra prerequisite is taken as a hypothesis. And the work is a formalisation: its novelty is in precision, verification, and counterexample, not in a new route through the mathematics.

## Acknowledgements

The preliminary results reported here were developed in the `fusion-category-lean` repository, whose `PLAN.md` records both the results and the retractions; the retractions are part of the evidence and are kept. The Haagerup–Izumi development in the sibling package is the work on which the feasibility argument rests.

## References

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# A Module inventory and dependency graph

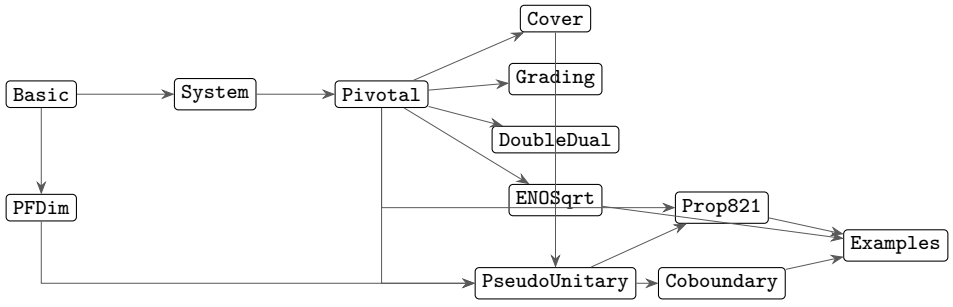



Figure 2: The import spine of the library (transitive imports suppressed; Sanity, Step4Checks and the individual Examples/ files are leaves off this spine). Examples/ is the umbrella over Fibonacci, Ising, Ostrik, $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$, Fibonacci–Ising and the step-4 checks; Yang–Lee and the pseudo-unitary instances are imported separately by the audit.

| Module | Lines | Decls | Content and status |
|---|---|---|---|
| Basic | 229 | 26 | fusion ring, fusion/FP matrices, IsPFEigenvector, FPData, global dimension. Complete; FPData is a hypothesis. |

| Module | Lines | Decls | Content and status |
|---|---|---|---|
| `System` | 451 | 19 | `FusionSystem`: $F$, `Finv`, snakes, $J$; $\mathrm{Vec}_{\mathbb{Z}/2}$ inhabitant with the 512-tuple pentagon. Complete. |
| `Pivotal` | 191 | 21 | `PivotalStructure`, `dim`, `IsSpherical`, `isSpherical_iff`, `pVal_dual`. Complete. |
| `PFDim` | 49 | 5 | `PFUnique`, $\varphi$ and its uniqueness. Complete (conditional on `FPData`). |
| `Cover` | 128 | 8 | `S4Data`, `CoverObj`, forgetful functor, `coverGlobalDim_double`. Scaffolding: no `PivotalStructure` on the cover yet. |
| `Grading` | 153 | 8 | universal grading, `pVal_dual`, two `def`s recording open targets. Formerly three axioms; now axiom-free. |
| `DoubleDual` | 155 | 9 | the older derived twist in the strict self-dual normalisation. Lags the $\psi$-derivation; to be rewritten against `Finv`. |
| `Coboundary` | 331 | 15 | `twistNorm`, `dim_eq_twistNorm`, `eno823_of_twistNorm` and its converse; the pointed `KappaMultiplicative` layer retained for reference. Complete. |
| `Prop821` | 474 | 21 | cyclic symmetry, `fpdim_char`, the weighted-norm bound, the equality case, `tMatrix`, `mult_iff_twist`. Complete modulo `DimEigen`. |
| `PseudoUnitary` | 452 | 37 | `ENO823`, `ENO823Exact`, `IsPseudoUnitary`, `IsPseudoUnitaryPivotal`, `eno823_of_pseudoUnitary`, `pivotal_ext`. Complete as a conditional reduction. |
| `ENOSqrt` | 232 | 13 | `SqrtData`, $b$, `Tmat`, $d$, `b_sq`, `trace_iff`, `toPivotal`. Complete; existence of $a$ is assumed. |
| `Sanity`, `Step4Checks` | 136, 147 | 8 | numerical and consistency checks. Complete; imported by the umbrella. |
| `Examples/FibonacciIsing` | 951 | 115 | Fibonacci and Ising over $\mathbb{R}$: every field proved, spherical, $\dim = \mathrm{FPdim}$. Complete; $\sim$100 s. |
| `Examples/YangLee` | 419 | 39 | the refutation of the un-hypothesised target. Complete. |
| `Examples/VecZ2Twisted` | 86 | 11 | two pivotal structures, $\dim g = \pm 1$. Complete. |
| `Examples/Ostrik` | 154 | 21 | the rank-three ring $K(1,0,1)$ and its `FPData`. Complete. |
| `Examples/`$\cdots$ | 638 | 38 | Fibonacci, Ising, pseudo-unitary instances, $Z/3$ note. |
| `MathlibPatches` | 55 | 0 | stubs for #42150 and the PF chain. Stub: WP0/WP5. |
| **Total** | **5,231** | **414** | 23 files; build 3,032 jobs; audit 73 declarations. |

# B The audit protocol

The protocol is a sequence of gates applied to every declaration before it is merged. Its purpose is to catch the failure modes F1–F4 of §6.7, none of which is caught by a successful build.

**Gate 1 — Inhabitation.** Every structure introduced gets a witness in the same commit, and the witness is a case from the literature or a standard construction, not a degenerate one. For `FusionSystem` the witness is $\mathrm{Vec}_{\mathbb{Z}/2}$; for `PivotalStructure` it is $p \equiv 1$ on a datum with $J \equiv 1$ and a datum without. A structure with no witness is not merged.

**Gate 2 — Sharpness.** Every hypothesis is either used essentially or is recorded as unused. A hypothesis $H$ is *used essentially* if dropping it makes the statement false on a witness. The two witnesses carried for this purpose are Yang–Lee (dropping pseudo-unitarity) and $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ (dropping multiplicativity). Where no counterexample is known, the declaration says so.

**Gate 3 — Correspondence.** Each theorem is entered in the statement-to-source table with (i) the source, (ii) the source's numbering in the edition used, (iii) the quantifiers and the carrier, (iv) any hypothesis of the source that is absent and why. The ENO numbering trap (8.23 in the

Annals, 8.22 on the arXiv) is recorded there. This is the gate a proof assistant cannot automate, and the only one that catches drift.

**Gate 4 — Axiom audit.** `AxiomAudit.lean` prints the axioms of every exported theorem. The build gate is:

```
cd FusionCategory
lake build
lake env lean AxiomAudit.lean > /tmp/audit.txt 2>&1
grep -v "propext, Classical.choice, Quot.sound" /tmp/audit.txt
```

A non-zero exit from the final `grep` usually means *no matches* — i.e. success — and must not be reported as a failure of Lean. Two failure modes are worth recording: a `#print axioms` applied to an `axiom` declaration itself errors, and a name whose module is not imported returns `Unknown constant`; both have been mistaken for real findings in this project.

**Gate 5 — Build hygiene.** No `sorry`, no `axiom`, no bare `module` header, warnings permitted. The full build must pass from a clean tree (WP0 Task 0.2), not merely incrementally.

**Gate 6 — Regression.** Each defect of Table 1 has a named test that fails if the defect returns: `VecZ2.system` (D1), the 512-tuple pentagon (D2), `VecZ2Twisted.system` (D3), `swapped_Finv_comp_ne_one` (D4).

## C The main statements

For reference, the target statements in mathematical form, with their Lean names.

**Pseudo-unitarity.** `IsPseudoUnitary S h` is the scalar equation

$$\sum_i \kappa_i \, \kappa_{i^*} \;=\; \sum_i f_i^2,$$

with no pivotal structure in it.

**The bare target.** `ENO823 S h` is

$$\exists\, P : \text{PivotalStructure } S, \quad P.\text{IsSpherical} \;\wedge\; \forall i,\ \dim_P(i) = f_i.$$

**The exact target.** `ENO823Exact S h` states: for every pseudo-unitary $S$ there is a unique pivotal structure $P$ with

$$P.\text{IsSpherical}, \qquad \forall i,\ \dim_P i > 0, \qquad \forall i,\ \dim_P i = f_i.$$

The existence is inside the unique existential; a rendering that lists a spherical structure and then asserts positivity and uniqueness *conditionally* on $\dim_P = f$ is strictly weaker, and the earlier draft of this appendix did exactly that. It is withdrawn.

**The conditional form already proved.** `IsPseudoUnitaryPivotal P` is the conjunction

$$(\forall i,\ \dim_P i > 0) \;\wedge\; \Big(\forall i, k,\ \sum_j N_{ij}^k \dim_P j = \dim_P i \cdot \dim_P k\Big),$$

and `eno823_of_pseudoUnitary` proves that these two conditions on one pivotal dimension, together with `PFUnique`, give `ENO823`.

**ENO's input.** `SqrtData` carries $a\colon \mathrm{Fin}\, n \to \Bbbk^\times$ with $a_i^2 = g_i$ and the monoidality of $a^2$ stated against $J^2$; then $b$, `Tmat` and $d$ are

$$b_c^{ij} = J_c^{ij}\frac{a_i a_j}{a_c}, \qquad T_c^{ij} = N_c^{ij} b_c^{ij}, \qquad d_i = a_i \,\mathrm{coev}_i\, \mathrm{ev}_{i^*},$$

and `trace_iff` states that $d_i d_j = \sum_c T_c^{ij} d_c$ for all $i, j$ is equivalent to (M) in the duality normalisation $\kappa_i' = \mathrm{coev}_i\, \mathrm{ev}_{i^*}$.

**The derived twist.** The object WP3 must construct is

$$J_{ab}^{c} \;=\; \frac{\psi_{a,b}|_c}{\psi_{b^*,a^*}|_{c^*}}, \qquad \psi_{a,b}\colon (a\otimes b)^* \to b^*\otimes a^*,$$

with $\psi$ the unique evaluation-intertwining isomorphism. The closed form in terms of $F$ and the restricted inverse block $M_b^{-1}$ that `PLAN.md` displays is *retracted there* and is not quoted as a target; WP3 Task 3.2 derives the closed form and checks it against Bartlett's Theorem 4.17.

**An ASCII rendering of the headline declaration.** The Unicode binders are spelled out; this is a rendering for readability, not the source, and it reproduces the existential conjunction of `ENO823Exact` verbatim.

```
-- the target, as the library actually defines it
def ENO823Exact (S : FusionSystem n R) (hFP : S.ring.FPData) : Prop :=
  IsPseudoUnitary S hFP ->
    exists! P : PivotalStructure S,
      P.IsSpherical /\
      (forall i, 0 < dim P i) /\
      (forall i, dim P i = hFP.fpdim i)
```

The conditional form `eno823_of_pseudoUnitary` replaces the antecedent by `IsPseudoUnitaryPivotal P` for a *given* $P$ together with `PFUnique`; it is weaker than `ENO823Exact` precisely in that it assumes a pivotal structure rather than producing one.

## D Changes in this revision

The first draft of this proposal was reviewed, and the review found eight substantive errors and a list of smaller ones. Each is recorded here with the change made, so that the revision can be checked against the review rather than against itself.

| | **Finding** | **Change** |
|---|---|---|
| 1 | The nontrivial double-dual permutation is based on an error: every simple is isomorphic to its double dual (ENO Prop. 2.1); pivotality is about the monoidal structure. | The permutation is removed. Remark 2.6 now separates the objectwise statement from the monoidal one; WP1 records the label question as closed; failure mode F3 no longer cites it. |
| 2 | The promised unconditional theorem exceeds the stated scope (multiplicities, the quadruple-dual theorem, complex scalars). | §2.6 states (S1)–(S4); $g$ is a hypothesis or Bartlett's Theorem 3.28; complex scalars are a WP1 task; the abstract says "multiplicity-free, characteristic zero". |

| | **Finding** | **Change** |
|---|---|---|
| 3 | The analytic package (old WP3, now WP4) misstates the phase correction: $\sigma$ is a tensor automorphism of the identity, $\tau$ is an ordinary natural automorphism, and $\sigma$ need not be sign-valued. | WP4 Task 4.5 rewritten with the two steps correctly typed, including the observation that a monoidal character leaves the defect $b$ unchanged; §4 and §4.5 corrected. |
| 4 | The square-root criterion for pivotality is invalid, and the cover is an equivariantization. | WP5 Task 5.3 withdraws the criterion and the no-2-torsion corollary, with the $\mathrm{Vec}_{\mathbb{Z}/2}$ counterexample; the cover is restated as an equivariantization. |
| 5 | The displayed exact target omits existence. | Appendix C reproduces `ENO823Exact` verbatim, including the existential conjunction, and records the weaker rendering as withdrawn; WP1 Task 1.5 makes this a proof obligation. |
| 6 | $X^2 = \mathbf{1} + 2X$ has no categorification over $\mathbb{C}$; a determinant or trace does not make the general twist scalar. | WP7 Task 7.3 turns the ring into a negative test citing [Ost02]; §6.4 states that the condition is an operator equality on each multiplicity space, with no scalar reduction. |
| 7 | The twist package (old WP2, now WP3) reproduces a retracted $\Omega$ formula, and the cyclic census is vacuous. | WP3 opens with the retraction, takes the closed form as a task rather than an input, and requires four genuinely non-trivial cocycles, kernel-checked; the carry-cocycle bug is recorded. |
| 8 | The claim about the *number* of pivotal structures is overstated. | The abstract, Theorem 4.5 and its discussion now claim only that the *dimension function* is not determined by the ring, and note that the set of structures is a torsor once one exists. |
| a | The triangle axiom is derivable, not independent. | WP1 Task 1.2 proves $t = t^2 \Rightarrow t = 1$ from the pentagon and invertibility. |
| b | The twist coherence needs the nonzero-entry qualification. | WP1 Task 1.3 states it, citing Bartlett's Corollary 3.26, and derives the F-free form only on the support. |
| c | (C)⇒(K) is not available in the signature. | Figure 1 redrawn: (C) and (M) are marked necessary and not sufficient, with the pointed false-$J$ datum as the witness; the missing implication is attributed to the reconstruction and the twist. |
| d | $\mathrm{FPdim}(\mathcal{C}) = \sum_i f_i^2$, without the extra square. | Corrected in §2.2. |
| e | The pseudo-unitarity definition is circular. | Definition 2.7 now uses ENO's squared norms, available before pivotality, with the equivalence to (5) as a lemma. |
| f | The cyclic symmetry is missing a dual. | Corrected to $N_{ij}^k = N_{jk^*}^{i^*}$, the `cyclic1` statement. |
| g | EGNO attribute the theorem to Proposition 9.5.1. | Corrected in Theorem 2.8 and §3, with EGNO's own note identifying it with ENO 8.23. |
| h | Higher indicators are cyclotomic integers, not integral by sphericity. | Corrected in §3. |